



OPEN Designing a residual-enhanced hybrid Prophet–LSTM framework for urban air pollution forecasting in Beijing

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Air pollution forecasting plays a vital role in mitigating the adverse impacts of deteriorating air quality on public health and urban sustainability. This study presents a data-driven hybrid framework for urban air quality forecasting, focusing on the accurate prediction of $PM_{2.5}$ concentrations. To ensure reliable model training, the Beijing Air Quality dataset was carefully preprocessed, including handling of missing values, removal of outliers, and feature selection based on correlation analysis. The proposed model introduces a residual-enhanced hybrid forecasting framework that integrates the statistical interpretability of Prophet with the nonlinear learning capacity of Long Short-Term Memory (LSTM) networks. Prophet is first employed to capture long-term trend and seasonality components of $PM_{2.5}$ time series, while an LSTM is trained on the residuals to model complex, nonlinear dependencies that Prophet cannot explain. Extensive experiments conducted on the air quality dataset demonstrate that the proposed Prophet+LSTM model significantly outperforms both traditional statistical methods and advanced deep learning baselines, achieving lower MAE and RMSE compared to AC-LSTM, LSTM-FC, CNN-LSTM, and XGBoost. These results highlight the effectiveness of preprocessing and hybrid modeling for accurate air quality forecasting and provide a framework for urban air quality monitoring.

Air pollution adversely affects human health, increasing both hospitalization and mortality rates. Consequently, the impact of polluted air on living organisms has remained one of the main subjects of scientific research for decades^{1–4}. Moreover, research in this area not only focuses on the nature of air pollution, but also on forecasting pollutant levels and developing strategies to mitigate their impacts^{5–8}.

Clean air is not merely a luxury but a fundamental necessity for human health and survival. However, individuals are often forced to remain in environments with poor air quality due to negligence or the inability to improve working conditions. Continuous exposure to poor air quality can progressively harm human health, ultimately leading to lasting and irreversible health complications. In addition, vulnerable populations such as infants and the elderly may experience more rapid and severe health consequences under poor air conditions. Long-term exposure to poor air quality has been associated with an increased incidence of various health conditions in later life, such as bronchitis, lung cancer, asthma, hypertension, arrhythmias, developmental abnormalities, allergic reactions, central nervous system disorders, cognitive decline, and both cardiovascular and respiratory impairments^{1,4}. Moreover, studies conducted by reputable institutions such as the U.S. Environmental Protection Agency and Health Canada have shown that approximately 68% of human diseases are associated with air pollution⁹. In addition, the study conducted in¹⁰ identifies air pollution as the second leading fatal risk factor.

The detrimental effects of poor air conditions tend to manifest gradually in the human body, often leading to their neglect or underestimation of their impact on health. This phenomenon is termed a latent effect, indicating that the harmful consequences of exposure do not manifest immediately but rather emerge over time¹¹. In addition to diminishing quality of life, poor air quality is a key factor that directly impacts psychological well-being and human productivity. The impact of air pollution extends to reduced labor productivity, diminished educational outcomes, impaired cognitive judgment, and has even been associated with elevated rates of criminal activity¹.

To effectively mitigate air pollution and improve air quality, it is essential to not only monitor pollution levels but also accurately identify the underlying causes and predict future concentrations. In this study, a hybrid forecasting framework is developed that leverages the interpretability of the statistical Prophet model and the

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nonlinear learning capacity of Long Short-Term Memory (LSTM) networks. First, Prophet is employed to capture the trend and seasonality components of $PM_{2.5}$ time series data, and then an LSTM model is trained on the residuals to model the complex nonlinear patterns not accounted for by Prophet. By combining these two complementary methodologies, the proposed residual-enhanced hybrid approach yields improved forecasting performance, offering a robust and interpretable solution for real-time air quality monitoring and policy planning in urban environments. The main contributions of this study are as follows:

- *Comprehensive preprocessing pipeline* This study introduces a fully integrated preprocessing framework for air quality prediction, encompassing missing value imputation, anomaly detection, label encoding, normalization, and data smoothing. The proposed pipeline enhances data reliability and model readiness, aspects often neglected in existing works.
- *Robust anomaly detection and data regularization* Anomaly detection and smoothing (regularization) are applied systematically before modeling, addressing one of the key challenges in time series datasets: noise and irregular sampling.
- *Incorporation of exogenous features (extra regressors)* The study explores the impact of external variables such as meteorological data and station information on model accuracy, providing a multi-dimensional perspective on air quality forecasting.
- *Hybrid use of statistical and deep learning models* A novel algorithmic design has been established by combining statistical approaches with deep learning architectures, leveraging the strengths of both paradigms.
- *Evaluation across multiple metrics* The predictive models are evaluated using diverse statistical metrics (e.g., MAE, RMSE), offering a holistic assessment of model performance.

The remainder of this paper is organized as follows. The “[Background](#)” provides detailed background information relevant to the problem. The “[Related work](#)” section provides an overview of previous research in the field of air quality forecasting. Section “[Dataset overview and preprocessing pipeline](#)” presents the dataset and the preprocessing pipeline applied to prepare the data for modeling. Section “[Residual-enhanced hybrid forecasting model: Prophet + LSTM](#)” introduces the proposed forecasting framework, highlighting the integration of Prophet and LSTM. Section “[Experimental results and discussion](#)” describes the experimental setup, evaluation metrics, and results obtained from comparative analyses. Finally, the “[Conclusion](#)” section concludes the paper and outlines potential directions for future research.

Background

In response to the increasing threat, many governments have introduced pollution-reducing regulations aimed at mitigating air pollution⁸. Although pollution mitigation measures such as emission control at coal-fired power plants, minimizing the use of hazardous fuels in residential areas and promoting cleaner alternatives, enforcing compliance in industrial facilities and vehicles, setting emission standards for cars and light trucks, optimizing energy structures and traffic systems, advancing clean environmental technologies, implementing emergency response plans for high-pollution episodes, and assigning legal responsibilities have achieved a certain level of success, $PM_{2.5}$ concentrations in many cities around the world still significantly exceed the WHO thresholds. WHO thresholds are shown in Table 1.

In the past, measuring and assessing air quality was both labor-intensive and costly. Devices such as gas sampling tubes, indicator papers, hygrometers, anemometers, particle counters, and gas detectors were commonly used to measure air quality. Today, advanced sensors, Internet of Things (IoT) devices, and continuous monitoring systems are employed to measure air quality. Such devices facilitate real-time data transmission and allow users to monitor air quality on a continuous and real-time basis.

Air pollution originates from diverse anthropogenic and natural sources, including industrial activities, vehicular emissions, and meteorological events. In this study, the focus is placed exclusively on outdoor air quality, particularly on the forecasting of $PM_{2.5}$ concentrations in urban environments, which represent a major indicator of environmental and public health impact.

When the concentration of volatile gas components responsible for air pollution exceeds certain thresholds, it can have detrimental effects on living organisms. In assessing air quality, the concentration of certain gaseous components and the amount of pollutants classified as particulate matter (PM) are taken into consideration. PM is a term used to describe all solid particles and liquid droplets suspended in the air, regardless of their shape, size, surface area, chemical composition, or source. These particles exhibit significant variations in terms of size, shape, and chemical composition. PM is considered a key indicator in air pollution assessments and can have serious adverse effects on human health. $PM_{2.5}$ refers to particulate matter with a diameter of 2.5 micrometers (μm) or smaller. These particles are approximately 1/30 the diameter of a human hair and are difficult to see with the naked eye. Due to their petite size, $PM_{2.5}$ particles can penetrate deep into the respiratory system, reaching the lungs and even entering the bloodstream. PM_{10} refers to particulate matter with a diameter of 10 micrometers (μm) or less. Although PM_{10} consists of particles larger than those in $PM_{2.5}$, they are still small enough to be invisible to the naked eye. These particles may deposit in the nasal cavity, pharynx, and upper airways. $PM_{2.5}$ is commonly associated with combustion-related sources, including motor vehicle emissions and industrial activities, whereas PM_{10} primarily originates from natural sources such as dust and pollen. In conclusion, $PM_{2.5}$ and PM_{10} are key indicators used in the assessment of air quality, and both pose risks to human health. Table 1 presents the major pollutants affecting air quality along with the safe limit values recommended by the WHO¹².

Deep learning (DL) is a subfield of machine learning that enables artificial neural networks to learn complex patterns from large datasets by utilizing multi-layered architectures. DL has demonstrated remarkable success in capturing nonlinear patterns, particularly in high-dimensional and large-scale datasets. In environmental data

Pollutant	Interim target level	Guideline value (µg/m ³ or mg/m ³)	Averaging time
PM _{2.5}	IT-1	35 µg/m ³	24-hour average
	IT-2	25 µg/m ³	24-hour average
	IT-3	15 µg/m ³	24-hour average
	AQG	15 µg/m ³ (24 h) / 5 µg/m ³ (annual)	24-hour / annual
PM ₁₀	IT-1	75 µg/m ³	24-hour average
	IT-2	50 µg/m ³	24-hour average
	IT-3	45 µg/m ³	24-hour average
	AQG	45 µg/m ³ (24 h) / 15 µg/m ³ (annual)	24-hour / annual
NO ₂ (Nitrogen Dioxide)	IT-1	40 µg/m ³	annual average
	AQG	25 µg/m ³ (24 h) / 10 µg/m ³ (annual)	24-hour / annual
O ₃ (Ozone)	IT-1	160 µg/m ³	8-hour average
	AQG	100 µg/m ³ (8 h)	8-hour average
SO ₂ (Sulfur Dioxide)	IT-1	125 µg/m ³	24-hour average
	AQG	40 µg/m ³ (24 h)	24-hour average
CO (Carbon Monoxide)	–	4 mg/m ³ (24 h)	24-hour average

Table 1. WHO Air Quality Guidelines (AQG, 2021)¹² and corresponding interim target (IT) levels with their concentration limits and averaging times. AQG represents the final guideline values recommended by the World Health Organization for optimal air quality, while IT-1 to IT-4 denote progressively stricter interim targets that countries can adopt as stepwise goals toward achieving the AQG standards.

analyses such as air quality forecasting, deep learning models can provide higher accuracy rates compared to traditional statistical methods^{13,14}. As an example, Zhang et al.¹³ effectively utilized Long Short-Term Memory (LSTM) networks to forecast the Air Quality Index (AQI), yielding promising results. Similarly, Li et al.¹⁴ employed a hybrid model combining Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks to capture temporal and spatial patterns in air pollution data, achieving effective predictions. These studies demonstrate that deep learning holds significant potential for air quality forecasting.

Related work

The association between air pollution and many of the aforementioned diseases is now well-established, highlighting the need for effective control measures. While the study presented in⁴ provides detailed data and observations regarding the strong correlation between air pollution and airway diseases, our study focuses on the prediction of air pollution. This study emphasizes the real-time identification and mitigation of poor air quality conditions, which were previously undetectable or difficult to manage, enabled by recent technological advancements. As a result, the related work is structured around prior research addressing the prediction and management of air pollution.

In their study, Long et al. proposed a smart Informer-based approach for long-term forecasting of indoor air quality⁹. The study highlights the adverse effects of indoor air pollution on human health and reveals the limitations of traditional forecasting methods. The proposed model captures temporal dependencies and relationships among data indicators through neural networks, and has demonstrated superior performance, particularly in scenarios with limited data points. Unlike the study by Long et al., which focuses exclusively on indoor environments and leverages an Informer model for sequence learning, our work addresses outdoor urban air quality forecasting, specifically targeting PM_{2.5} concentrations using a hybrid sequential Prophet–LSTM framework. While both approaches incorporate temporal dependency modeling, our method integrates Prophet’s decomposable trend modeling with LSTM’s nonlinear residual learning, enabling it to effectively capture both deterministic trend–seasonality patterns and stochastic fluctuations.

Moreover, Monteil et al.¹⁵ address the integration of differential equation solvers with deep learning methods to enhance the computational efficiency and geographic scalability of air pollution forecasting models. Although traditional pollution forecasting methods are typically based on partial differential equations (PDEs), their scalability is limited due to high computational costs and challenges in parameterization. In the study, a deep learning-based surrogate model is developed using large datasets generated by PDE-based air pollution models, and it is shown to be successfully applicable across different geographic regions. Tests conducted in the city of Dublin reveal that the proposed deep learning model operates twice as fast as traditional PDE solvers while maintaining spatial consistency and improving prediction accuracy. This approach reduces the computational burden of air pollution modeling and has the potential to make large-scale and high-resolution forecasts more accessible. While Monteil et al. focused on coupling deep learning with physics-based PDE solvers to improve spatial scalability and computational efficiency, our study follows a data-driven hybrid forecasting paradigm without relying on physical model simulations. Instead of surrogate modeling for PDE-based systems, we employ a sequential Prophet–LSTM hybrid that directly learns from observed temporal sensor data, emphasizing trend decomposition and residual correction. This enables our framework to capture both long-term non-stationary

dynamics and short-term fluctuations in urban PM_{2.5} levels, offering a practical alternative for real-world forecasting where physical model parameters are unavailable or difficult to estimate.

In another study, Ahn et al.⁶ propose yet another deep learning-based approach for air quality prediction. Emphasizing the limitations of traditional threshold-based control systems, the study employs Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks to more effectively model time-dependent variables. In the study, time series forecasting is performed using sensor data that include various factors such as CO₂, fine particulate matter, temperature, humidity, light intensity, and volatile organic compounds. The GRU model has demonstrated superior performance in terms of accuracy compared to both the LSTM and linear regression methods. Additionally, the study aims to enhance the model's predictive power through an algorithm designed to determine the optimal time step. Unlike Ahn et al., whose work focuses on indoor air quality prediction and evaluates stand-alone recurrent architectures (LSTM and GRU), our framework targets outdoor PM_{2.5} forecasting. While Ahn et al. emphasize optimizing time-step selection to improve temporal modeling, our approach enhances performance by combining Prophet's trend decomposition with LSTM's residual learning, achieving improved prediction quality across urban outdoor conditions.

The performance of a single predictive model may be constrained by its algorithmic design or structural framework, hindering its adaptability to intricate data relationships. For this reason, several researchers have developed hybrid modeling approaches to enhance the predictive accuracy of forecasting systems^{16,17}. Wang et al.¹⁶ propose a dual-layer time series forecasting model that combines LSTM and GRU architectures for PM_{2.5} prediction within an IoT-enabled air pollution monitoring system. It has been noted in the literature that conventional approaches struggle to model the nonlinear characteristics of smog data. To overcome this challenge, LSTM's strength in capturing long-term dependencies is integrated with GRU's efficient learning capability. The developed model has been trained using both national environmental monitoring data and data collected through IoT sensor nodes. The study serves as a strong example demonstrating the potential of IoT-based dual-layer RNN models in air pollution monitoring and early warning systems. Compared to Wang et al., whose IoT-based dual-layer RNN focuses on real-time smog prediction using combined LSTM and GRU layers, our study adopts a hybrid Prophet–LSTM framework that emphasizes trend decomposition and residual learning for outdoor PM_{2.5} forecasting. While both models address nonlinear temporal dependencies, our approach combines the complementary strengths of Prophet and LSTM to better capture short-term fluctuations in outdoor air quality parameters, thereby improving overall forecasting performance.

Researchers in article¹⁸ developed and evaluated deep learning-based spatio-temporal models for PM_{2.5} forecasting in urban environments, addressing the challenge of accurately predicting fine particulate matter concentrations. Using a large multivariate dataset from Beijing they compared the performance of LSTM, Bi-LSTM, GRU, Bi-GRU, CNN, CNN-GRU, and a proposed hybrid CNN-LSTM model. The methodology involved rigorous preprocessing steps, including missing value imputation, categorical encoding, normalization, correlation-based feature selection, and the integration of spatial dependencies through adjacent station data. Experimental results, assessed via MAE, RMSE, and R^2 across multiple batch sizes and forecast horizons, demonstrated that the hybrid CNN-LSTM consistently outperformed all baseline models by combining CNN layers for spatial feature extraction with LSTM layers for modeling temporal dependencies. This approach was shown to be robust for short-term air quality prediction and has potential for broader application in smart city contexts. While Bekkar et al. demonstrated the strong performance of a CNN–LSTM hybrid through spatial-temporal feature extraction, their approach mainly targeted short-term PM_{2.5} prediction within a deep neural network framework. In contrast, our Prophet–LSTM hybrid focuses on residual-based sequential learning, allowing effective long-term forecasting with enhanced interpretability. By integrating trend decomposition with nonlinear residual modeling, our framework provides a simpler yet explainable alternative that performs robustly without requiring complex convolutional architectures or multi-station spatial inputs.

In another hybrid model, Huang et al.¹⁷ present a backpropagation (BP) neural network optimized using an improved particle swarm optimization (PSO) algorithm for forecasting urban air quality. The study addresses the common weaknesses of traditional BP networks in the literature, such as slow learning and entrapment in local minima, by enhancing the PSO algorithm's inertia weight and learning factors to overcome these limitations. An adaptive mutation strategy was employed to prevent the algorithm from getting stuck in local optima. The model conducted AQI prediction by leveraging multi-pollutant data (PM_{2.5}, PM₁₀, NO₂, CO, SO₂, and O₃) collected in Chongqing over the 2018–2019 period. The experimental findings demonstrate that the enhanced PSO-BP network outperforms both standard PSO-BP and multivariate regression models, yielding lower prediction errors and an accuracy exceeding 99%. While Huang et al. enhanced a BP neural network through an improved PSO algorithm to optimize convergence and local search performance, our study adopts a hybrid framework emphasizing trend decomposition and residual learning rather than metaheuristic weight optimization. Unlike the PSO-BP model, which focuses on algorithmic optimization of network parameters, our approach aims to deliver accurate and stable long-term PM_{2.5} forecasts under dynamic urban conditions.

In recent years, statistical time series models have also become a frequent subject of study^{19,20}. Among these models, Autoregressive Integrated Moving Average (ARIMA), Seasonal Autoregressive Integrated Moving (SARIMA), and Prophet are well-known, ready-to-use forecasting tools. ARIMA and its seasonal variant, the SARIMA, have long been fundamental tools in time series forecasting due to their interpretability and solid theoretical foundation. The ARIMA model captures linear dependencies in time series data by combining autoregressive (AR), differencing (I), and moving average (MA) components. It is suitable for stationary or transformed-to-stationary series and has been widely used in environmental and air quality forecasting. In response to the limitations of classical statistical models, Facebook has developed Prophet, a decomposable time series forecasting model based on an additive framework. Prophet models the data as a sum of trend, seasonality, and holiday effects, and is robust to missing data, outliers, and non-linear trends. While traditional statistical models such as ARIMA, SARIMA, and Prophet offer interpretability and robustness for linear or

quasi-linear time series, they often fail to capture complex non-linear dependencies inherent in air quality data. In our study, Prophet is integrated with an LSTM network in a residual-enhanced hybrid framework, allowing the model to retain Prophet's decomposable structure while leveraging LSTM's strength in learning non-linear residual components. This sequential integration enables more accurate and adaptive forecasting of PM_{2.5} concentrations under dynamic urban conditions compared to purely statistical approaches.

In addition, deep learning-based forecasting studies have increasingly emphasized the importance of modeling spatio-temporal dependencies, as many real-world phenomena depend not only on temporal patterns but also on spatial interactions. Traditional time-series approaches that consider only temporal correlations are often insufficient when the target variable is influenced by multiple spatially distributed sources. This dual dependency substantially increases the complexity of the prediction problem, requiring models capable of jointly capturing temporal evolution and spatial heterogeneity. This challenge has also been addressed in recent works^{21,22}. In our study, this complexity arises because the predicted pollutant levels depend on both time and monitoring location, and the proposed hybrid framework is designed to manage these intertwined spatial and temporal dynamics effectively.

Air pollution forecasting has been extensively studied for decades and remains a vital area of research due to its ongoing relevance and societal impact. Recently, air pollution forecasting has increasingly employed deep learning models²³, especially LSTM^{16,24,25} and Transformer-based approaches^{26,27}, in parallel with traditional machine learning techniques^{28,29} due to their strong performance on time series data.

Dataset overview and preprocessing pipeline

The data collected from environments through sensors can be categorized as time series data. When working with time series data, it is important to recognize that earlier data points are correlated with later ones. In other words, past observations influence future values, and therefore, the temporal order of the data must be preserved. The temporal correlation in time series data must be carefully considered when splitting the data into training, validation, and test sets.

As shown in Fig. 1, a machine learning model traditionally comprises three main stages: (i) data preprocessing, (ii) model inference, and (iii) performance evaluation. In digital environments, data are collected from various sources and in different formats. These raw data are often unsuitable for direct analysis due to issues like incomplete records, outlier presence, inconsistent formats, and varying feature ranges. The shortcomings in the dataset can compromise the generalization capability of the employed machine learning algorithm. Therefore, carefully analyzing the characteristics of the data and applying appropriate preprocessing steps can significantly enhance the performance of the machine learning model.

To provide a clearer understanding of the preprocessing steps applied to the dataset, the subsequent section presents details about the dataset and its characteristics.

Dataset

In this study, the Beijing Multi-Site Air Quality Data Set³⁰, a publicly available time series dataset, was employed for air quality forecasting tasks³¹. The Beijing Air Quality dataset is a time-dependent dataset that includes air pollution levels and meteorological conditions in Beijing, the capital of China. This dataset covers a four-year period from 2013 to 2017 and contains hourly data collected from a total of 12 different air quality monitoring stations. The dataset consists of 420,768 records and 18 attributes. Table 2 presents the attributes of the dataset along with their characteristics and statistical information. Although the statistical information for date-related columns may not be meaningful, they were retained in the table to provide a complete overview of the dataset.

The time-related variables (year, month, day, and hour) serve as the foundation for time series analysis. Hourly data, in particular, are well-suited for fine-grained time interval forecasting models such as Prophet and LSTM. The similarity between the mean and median values implies a symmetric distribution and reflects a uniform temporal distribution of the data.

The weather-related variables (temperature (TEMP), pressure (PRES), dew point temperature (DEW), mean wind speed (WSPM), and wind direction (wd)) are key meteorological factors that directly influence PM_{2.5} levels. The missing data rates are very low (<1%), indicating the reliability of these variables.

PM_{2.5} is one of the most critical indicators of air pollution in terms of its impact on human health. The World Health Organization (WHO) recommends an annual mean PM_{2.5} concentration limit of 10–15 µg/m³¹². The missing data rate is relatively low at 2.08% and does not compromise the statistical validity of the dataset. Since the median is lower than the mean, the distribution may be right-skewed. As can be seen in Fig. 2, the records exhibit a right-skewed frequency distribution for the PM_{2.5} attribute. A large number of observations fall within the 0–75 µg/m³ range, indicating that PM_{2.5} levels mostly remained within this interval. In contrast, values of 500 µg/m³ and above, although rare, are observed and represent alarming levels for severely polluted days. Figure 2 provides important insights into the distribution of air pollution concentrations and the occurrence rate of critical pollution levels.

Data preprocessing

Data preprocessing is an essential step in machine learning, as the success of a model highly depends on the quality of the data used. Data preprocessing is intended to refine raw data into a format that facilitates easier handling and improved performance by predictive models. Missing value imputation, anomaly detection and removal, label encoding, normalization, and data regularization are preprocessing steps employed to improve data quality and boost the effectiveness of machine learning models.

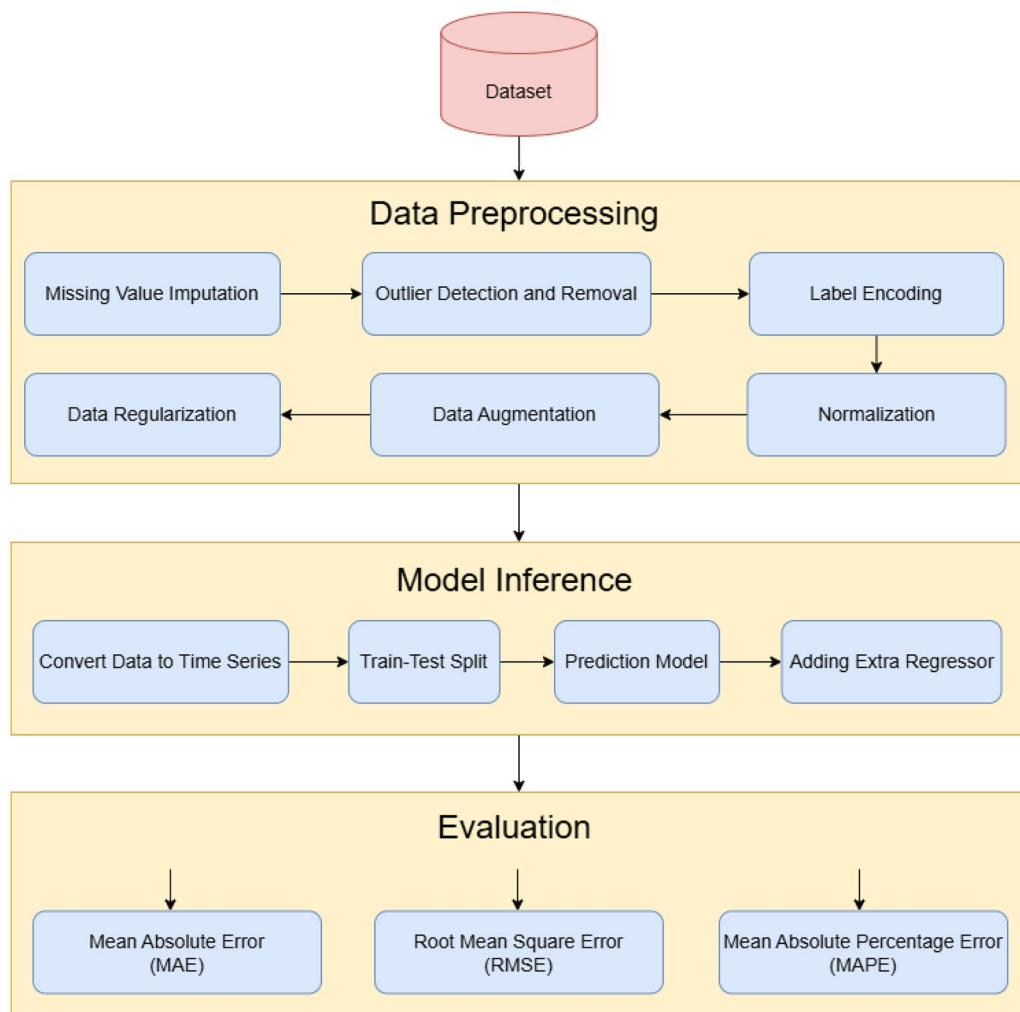


Fig. 1. Schematic representation of the proposed air quality prediction pipeline.

Feature	Type	Min	Max	Mean	Median	Std. Dev.
Year	int	2013	2017	2014.66	2015	1.1771
Month	int	1	12	6.52	7	3.4487
Day	int	1	31	15.72	16	8.8001
Hour	int	0	23	11.5	11.5	6.9221
PM _{2.5}	float	2	999	79.7934	55.0	80.8223
PM ₁₀	float	2	999	104.6026	82.0	91.7724
SO ₂	float	0.2856	500	15.8308	7.0	21.6506
NO ₂	float	1.0256	290	50.6385	43.0	35.1279
CO	float	100	10000	1230.7664	900.0	1160.1827
O ₃	float	0.2142	1071	57.3722	45.0	56.6616
TEMP	float	−19.9	41.6	13.5389	14.5	11.4361
PRES	float	982.4	1042.8	1010.7469	1010.4	10.4740
DEWP	float	−43.4	29.1	2.4908	3.1	13.7938
RAIN	float	0	72.5	0.0644	0	0.8210
wd	categorical	–	–	–	–	–
WSPM	float	0	13.2	1.7297	1.4	1.2463
station	categorical	–	–	–	–	–

Table 2. Overview of Beijing Air Quality Dataset features and statistical summaries.

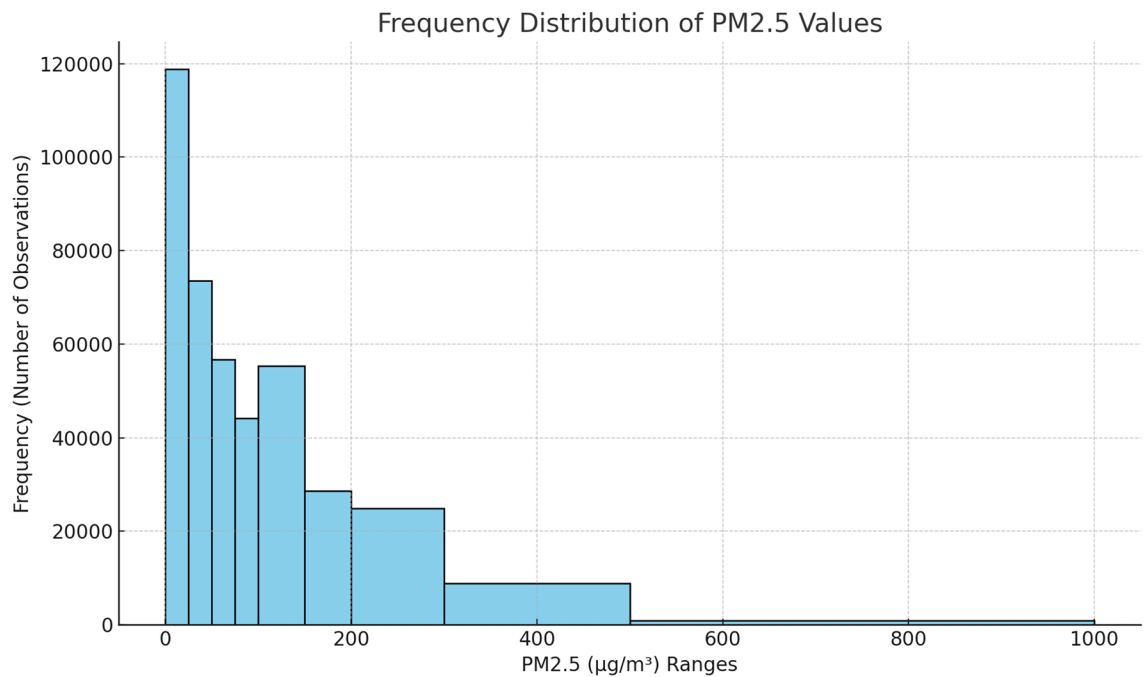


Fig. 2. Distribution of PM_{2.5} concentration levels observed in the Beijing Air Quality dataset.

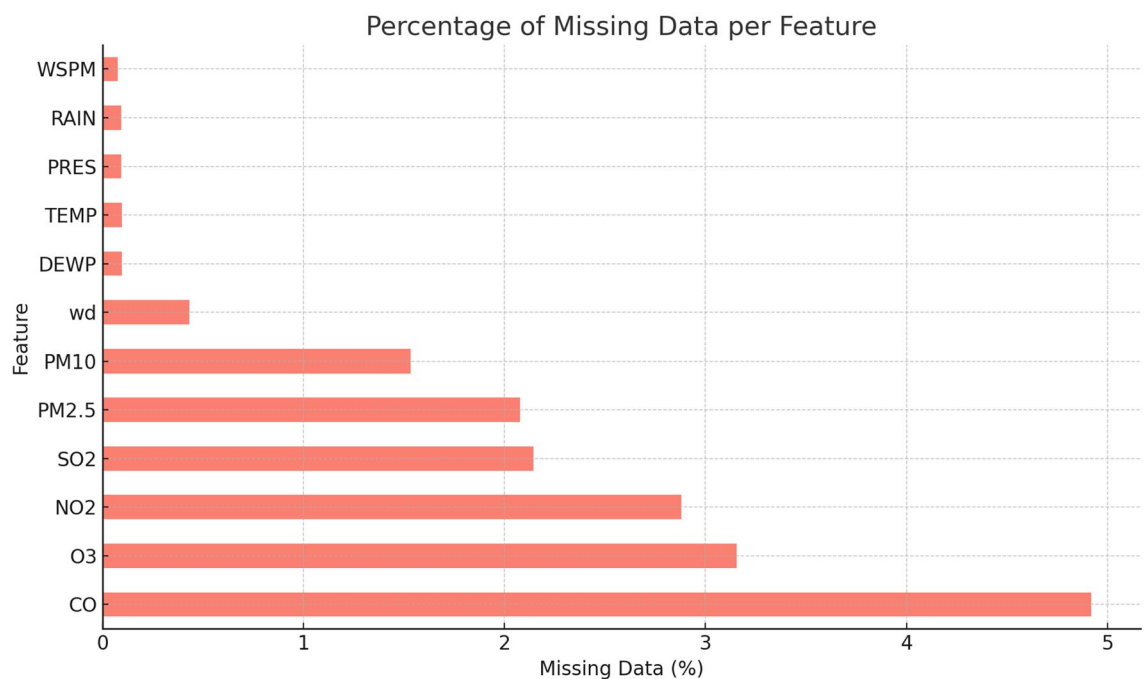


Fig. 3. Percentage of missing values for each feature in the Beijing Air Quality dataset.

Missing value imputation

The presence of missing data is a prevalent and natural occurrence in datasets, which may significantly impair the generalization capability of machine learning models and thereby reduce their overall performance. Including missing data in the analysis without proper treatment can lead to incorrect model learning or the exclusion of certain samples from the training process altogether. Moreover, some algorithms and models have low tolerance to missing data and may not function properly or at all when such data are present. Hence, addressing missing values using suitable methods is crucial for model performance. The dataset utilized in this research exhibits missing values, which are clearly demonstrated in Fig. 3.

Techniques for handling missing data can differ according to the data type and its underlying distribution. Data imputation techniques can be categorized into statistical and machine learning-based methods³². Widely used imputation techniques range from basic statistical strategies such as replacing missing values with the mean, median, or mode to more complex approaches like regression-based predictions, multiple imputation frameworks, KNN-based estimation, and interpolation methods that rely on temporal or spatial continuity in the data. For time series data, interpolation methods, including linear and spline approaches, provide a practical means of estimating missing observations by leveraging temporal continuity. Selecting an imputation method that aligns well with the underlying structure of the dataset can significantly enhance the model's overall performance and reliability.

Time series consist of time-dependent observations measured at regular intervals. Each observation is naturally contextualized by the preceding and succeeding observations. Therefore, A missing data point can be reasonably inferred based on the temporal continuity provided by its neighboring values. Time series data often follow smooth trends with incremental changes, rather than sudden fluctuations, making interpolation-based techniques well-suited for imputing missing values. In light of the aforementioned reasons, the proposed model adopts linear interpolation for imputing missing data.

In linear interpolation, the unknown value y at a given position x , located between two known points, is calculated as follows:

$$y = y_1 + \frac{(x - x_1)}{(x_2 - x_1)} \cdot (y_2 - y_1) \tag{1}$$

In Eq. (1), (x_1, y_1) and (x_2, y_2) represent the two known data points between which interpolation is applied. The variable x denotes the position of the unknown value to be estimated, while y refers to the interpolated value corresponding to that position.

Anomaly detection and removal

In data analysis and machine learning applications, outliers are defined as observations that significantly deviate from the general distribution of the dataset^{33,34}. Such deviations can result from measurement errors, data entry mistakes, device malfunctions, or rare real-world events. Detecting and appropriately handling outliers is a crucial preprocessing step that directly influences model performance, as outliers can undermine the generalization ability of models, particularly in regression and distance-based classification algorithms, and lead to misleading outcomes. The elimination of outliers from the dataset significantly contributes to the improvement of machine learning models by strengthening their ability to generalize to unseen data.

A dataset may not contain outliers, or if present, their occurrence may be so minimal that it can be deemed statistically insignificant. Therefore, outlier removal is not always a necessary step. Prior to conducting any analysis, examining the entries contained in the dataset is essential. The appropriateness of outlier elimination should be assessed based on a comprehensive understanding of the dataset structure, the modeling framework, and the inherent complexity of the target problem. Eliminating outliers can be problematic, especially when they characterize the nature of the problem, when the dataset is imbalanced, or when the number of samples is limited. Furthermore, when the employed model is already tolerant to outliers, their removal may not be warranted.

At this point, it becomes essential to address the following question: Should outliers be removed in the context of this study? It would be appropriate to answer this question based on the statistical characteristics and metadata of the dataset. High standard deviation, a considerable difference between maximum and minimum values, or significant skewness in the dataset can serve as indicators of potential outliers.

Table 3 displays the descriptive statistics of the PM_{2.5} feature from the employed dataset. The PM_{2.5} values range between 2 and 999 $\mu\text{g}/\text{m}^3$, with a mean of approximately 79.84 and a standard deviation of 80.95, indicating a high variance and the possible presence of extreme values. The distribution is right-skewed, as the mean exceeds the median (55), and further inspection reveals potential outliers, particularly values near the upper bound.

The statistical metrics shown in Table 3 explicitly reveal the existence of anomalous or outlier values within the dataset. Moreover, the same inference can be readily made upon analyzing Figs. 4 and 5. Figure 4 demonstrates the Z-score distribution of PM_{2.5} values, revealing a significant concentration near the mean and a heavy tail on the right, where values exceeding the +3 threshold are considered statistical outliers. Figure 5 further illustrates the impact of these outliers on the overall data distribution. After removing such extreme values, the resulting

Count	420768
Mean	79.839
Std.	80.950
Min	2.000
25%	20.000
50%	55.000
75%	111.000
Max	999.000

Table 3. Descriptive statistics of PM_{2.5} concentrations in the dataset.

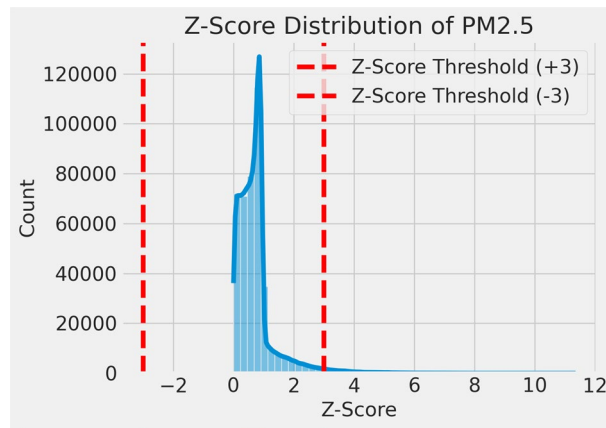


Fig. 4. Z-score distribution of $PM_{2.5}$.

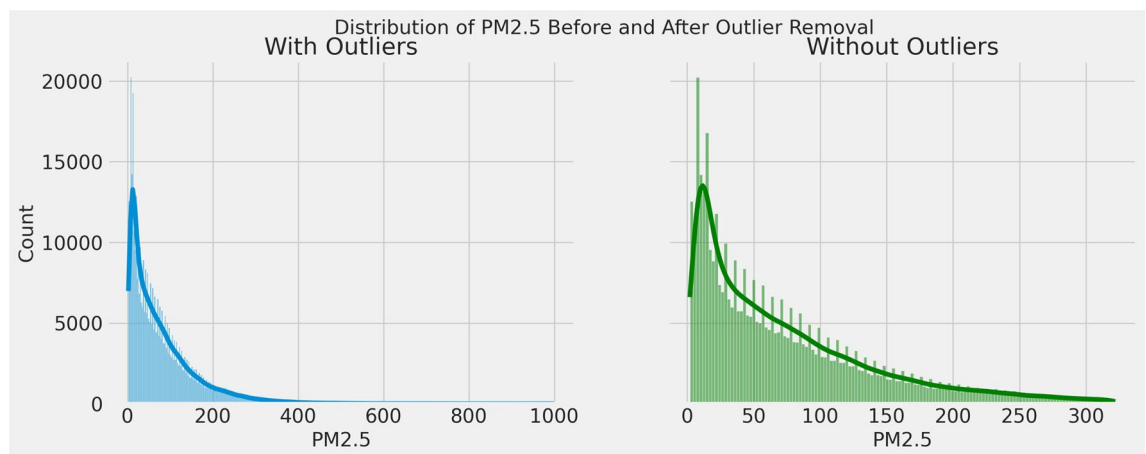


Fig. 5. Distribution of $PM_{2.5}$ before and after outlier removal.

distribution appears more symmetrical and compact, which facilitates more robust and stable model training. Hence, outlier detection and removal are crucial in ensuring data reliability and preventing model bias due to anomalous observations.

The visual comparison clearly illustrates that the presence of outliers skews the $PM_{2.5}$ distribution and affects statistical balance. By applying Z-score filtering, a more representative and compact distribution is achieved, reinforcing the importance of outlier removal prior to model training in standard forecasting tasks. The Z-score method was selected for outlier detection due to its simplicity, interpretability, and suitability for large datasets³⁵. Although the $PM_{2.5}$ distribution exhibits a slight positive skew, the Z-score approach was applied here not as a strict normality-based filter, but as a controlled noise removal mechanism to eliminate implausible sensor readings and data transmission errors. The threshold was carefully calibrated to preserve genuine high-concentration pollution events while discarding only unrealistic outliers. As a result of this procedure, a total of 7,637 outliers (1.815% of the dataset) were removed, ensuring that the integrity of significant pollution peaks remained unaffected.

$$Z = \frac{x - \mu}{\sigma} \quad (2)$$

Here, x denotes the observed data point, μ represents the mean of the dataset, and σ corresponds to the standard deviation.

Label encoding

Categorical variables provide insights into an observation's classification, category, grouping, or qualitative characteristics. For instance, in an air quality forecasting model, variables such as city, station type, or season may play a decisive role in predicting $PM_{2.5}$ concentrations. Excluding such variables not only reduces the explanatory power of the model but also risks overlooking important correlations and causal relationships.

Given the reasons outlined above, non-numerical features should be appropriately integrated into the model. This procedure is commonly implemented through label encoding, one-hot encoding, or embedding techniques. In this way, the model becomes capable of processing categorical information through numerical representations, enabling more comprehensive and accurate predictions^{36,37}.

The dataset contains two categorical features: wind direction and station. Neither of these categorical variables possesses any inherent ordering, hierarchy, or significance among their respective categories. Therefore, to avoid introducing artificial ordering or weighting into the feature space, one-hot encoding was preferred over direct label encoding.

Figure 6 shows the wind direction (wd) column in the Beijing Air Quality dataset, which represents wind direction values using standard compass notations, including cardinal (e.g., N, S, E, W), intercardinal (e.g., NE, SW), and secondary intercardinal directions (e.g., NNE, SSW). These 16-point compass directions are commonly used in meteorological datasets to indicate atmospheric wind flow.

The station feature represents twelve distinct air quality monitoring locations across Beijing, including urban (e.g., Dongsi, Tiantan), suburban (e.g., Shunyi, Changping), and rural sites (e.g., Huairou, Dingling). These spatial differences allow the dataset to capture geographic variability in PM_{2.5} concentration levels and facilitate spatially informed forecasting.

As a result of the applied one-hot encoding, 16 additional features were generated from the wd column and 12 from the station column, yielding a total of 28 new features appended to the original dataset. Consequently, the dataset was transformed into a 44-dimensional feature space.

In this study, both station and wd features were encoded using one-hot encoding, resulting in an expanded but more informative feature space. Although this approach increases the dimensionality of the dataset, it prevents the model from learning irrelevant ordinal relationships between non-numeric categories, which can occur with label encoding.

Normalization

Normalization enables the comparison and joint use of values originating from different distributions or measured in distinct units by transforming them into a scale that is independent of original measurement units and distribution-specific characteristics^{38,39}.

In the absence of normalization, features with larger numerical scales can dominate the learning process, leading to biased models, unstable training, and decreased generalization performance, particularly in algorithms that are sensitive to distance or gradient magnitude⁴⁰. Moreover, in deep learning models, unnormalized inputs may result in exploding or vanishing gradients during backpropagation, severely impairing the convergence and stability of the learning process.

To ensure that all numerical features contribute equally to the model, Min-Max normalization was applied using Eq. (3). As illustrated in Fig. 7, this transformation rescales each feature into a [0, 1] interval, preserving the original distribution while eliminating scale disparities. This is particularly beneficial in distance-based and gradient-sensitive models.

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

(3)

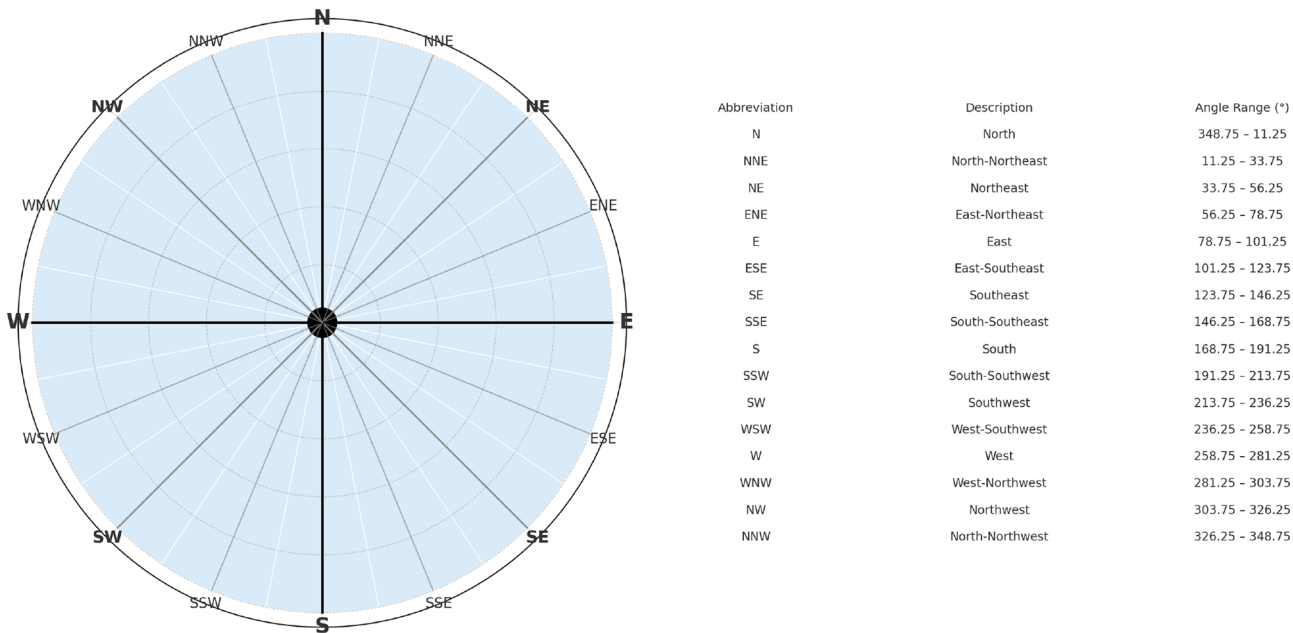


Fig. 6. Wind direction (wd) feature in the Beijing Air Quality dataset, represented using the 16-point compass rose system, with corresponding directional names and angular ranges.

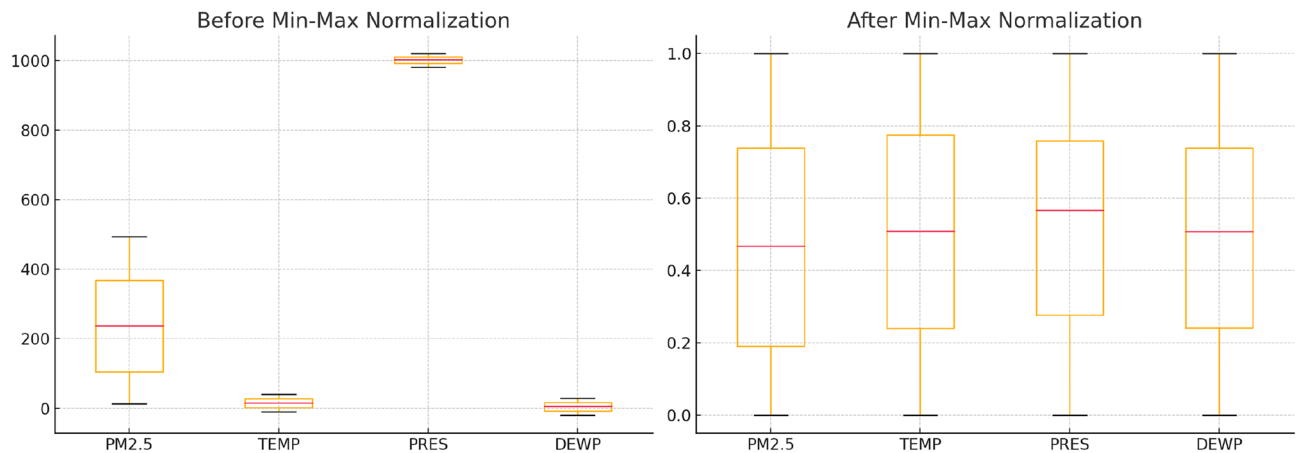


Fig. 7. Boxplot visualization of selected features before and after Min–Max normalization.

where x denotes the original value of a given feature, $x_{\min}$ and $x_{\max}$ represent the minimum and maximum values of that feature within the dataset, respectively, and x' corresponds to the normalized value scaled to the interval $[0, 1]$.

In Fig. 7, the left panel shows the original distribution of selected features with varying numerical ranges. The right panel demonstrates how all features are rescaled into the $[0, 1]$ interval using Min–Max normalization, ensuring comparable scales across variables while preserving their internal distribution. This figure serves as a conceptual example and does not depict actual data from the study.

Data augmentation techniques were not applied in this study, as the dataset was sufficiently large, diverse, and temporally rich to ensure robust model training without synthetic expansion.

Data regularization

At first glance, outlier detection and data regularization (or smoothing) may seem alike; however, they are different concepts. Therefore, it is essential to resolve the conceptual ambiguity between the two terms before proceeding with any further analysis. Both approaches aim to enhance data stability and improve the model's generalization capability; however, they differ significantly in their methodological and practical aspects.

Outlier detection involves identifying and handling unusual observations within a dataset that deviate statistically from the overall distribution. The goal is to detect and manage extreme values, which can arise from various factors, to avoid negative effects on the model⁴¹. On the other hand, data regularization refers to a set of processes aimed at reducing noise and suppressing excessive volatility in the data to achieve a smoother and more predictable structure^{42,43}. The objective is to render the data with stable variance, enhanced interpretability, and improved suitability for modeling. The two methods serve complementary roles in preprocessing pipelines, and their combined or selective application depends on the nature of the data and the modeling objective.

Figure 8 presents the moving average smoothing in a two-panel visualization to illustrate its effect on the $PM_{2.5}$ signal. The upper panel displays the complete normalized $PM_{2.5}$ series together with the 100-hour moving average, where the light blue region denotes the original data and the red curve represents the smoothed trend. The shaded grey area and dashed vertical lines indicate the 30-day segment selected for detailed inspection. The lower panel zooms into this specific window, clearly showing how the moving average suppresses short-term fluctuations while preserving the underlying trend, thereby enhancing the interpretability of temporal dynamics within the dataset.

The moving average is defined as:

$$\bar{x}_t = \frac{1}{2k+1} \sum_{i=-k}^k \bar{x}_{t+i} \quad (4)$$

where $\bar{x}_t$ is the smoothed value at time t , and $2k+1$ is the window size (in this case, 100 hours).

This technique is especially useful in time-series analysis when the data is highly volatile or noisy, as it enhances interpretability by reducing short-term irregularities that may not reflect the underlying trend.

Residual-enhanced hybrid forecasting model: Prophet + LSTM

In time series forecasting, a variety of models from different domains are employed. These algorithms can be categorized into four main groups: (i) statistical models^{19,20,44}, (ii) traditional machine learning models^{28,29,43}, (iii) deep learning models^{5,8,24}, and (iv) hybrid models^{7,13,27}. A review of existing studies reveals that hybridization is commonly performed either between deep learning models and traditional machine learning models or among different deep learning models themselves.

Prophet effectively captures the deterministic components of air quality dynamics (namely long-term trends and seasonal variations) through its additive decomposable structure. However, empirical observations indicated

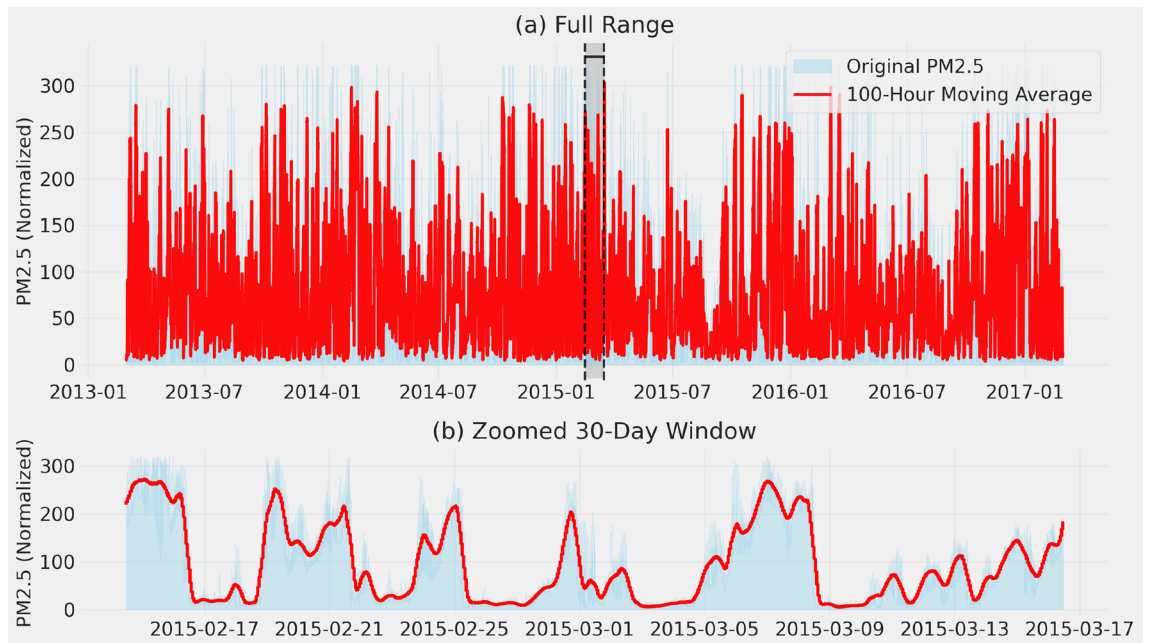


Fig. 8. Visualization of $PM_{2.5}$ moving average smoothing. The upper panel shows the entire normalized $PM_{2.5}$ series, while the lower panel presents a zoomed 30-day segment to clearly illustrate the smoothing effect.

that the model's predictions tend to stay close to the mean $PM_{2.5}$ levels, limiting its ability to capture nonlinear temporal deviations. To address this limitation, the residual errors produced by Prophet are modeled using an LSTM network, which is well-suited for learning complex, time-dependent nonlinear patterns. In this hybrid formulation, the LSTM learns and compensates for the temporal dependencies that Prophet fails to represent, thereby enhancing the model's overall forecasting accuracy. Such additive integration of statistical and neural components has been widely adopted in the literature.

Convert data to time series

Facebook Prophet is a model designed for time series forecasting and expects input data to conform to a specific format. Prophet's forecasting engine operates on a data structure containing only two columns: the *ds* column, which represents the time component, and the *y* column, which denotes the target variable to be predicted. As any data input deviating from this structure may impede the model's operation, it is essential to transform the raw data into the required format.

The Beijing dataset used in this study initially contains time components separated into year, month, day, and hour. To ensure compatibility with Prophet, these columns were first combined to produce a single timestamp, as depicted below.

$$\begin{aligned} \phi: \mathbb{R}^4 &\rightarrow T \\ d_{s_t} &= \phi(\text{year}_t, \text{month}_t, \text{day}_t, \text{hour}_t) \end{aligned} \quad (5)$$

Here, t denotes the observation index of the time series. As a result of this step, an hourly-resolution timestamp (*ds*) was generated for each observation. In the subsequent step, the values corresponding to the target variable, $PM_{2.5}$ concentration, are to be provided as input to Prophet.

As a result, a two-column format (d_{s_t}, y_t) suitable for direct processing by Prophet was obtained. Using this format, the Prophet model was trained and performed forecasts on hourly $PM_{2.5}$ concentrations. This transformation constitutes a critical preprocessing step that supports the model's learning by preserving data integrity and the inherent time series structure.

Dataset splitting

During the training and validation of time series models, it is imperative to employ temporally constrained partitioning techniques rather than conventional random sampling to preserve temporal dependencies. The underlying justification is the preservation of causality in temporally correlated datasets, ensuring that model inference strictly adheres to the paradigm of forecasting future outcomes from prior observations. In this context, to evaluate the performance of the Prophet model, the dataset was split into 80% for training and 20% for testing, while preserving the temporal order.

In this study, the dataset was divided into 80% for training and 20% for testing to ensure a sufficient historical window for learning long-term temporal dependencies, particularly for the LSTM component. Moreover, this

splitting method enables Prophet to observe and learn complete seasonal cycles, thereby supporting its ability to effectively model both annual and weekly seasonality components. This approach also allows for a realistic evaluation of the model's performance and ensures that the dataset's full potential is utilized to the greatest extent possible.

Let N represent the total number of observations in the dataset; the training portion is designated as the first $0.8 \times N$ instances, according to the following formal definition:

$$\text{Train Set} = \{(d_{s_t}, y_t) \mid t = 1, 2, \dots, 0.8 \times N\} \quad (6)$$

The remaining $0.2 \times N$ observations constituted the test dataset:

$$\text{Test Set} = \{(d_{s_t}, y_t) \mid t = 0.8 \times N + 1, \dots, N\} \quad (7)$$

Proposed hybrid model

Facebook Prophet model

The Prophet model, developed by Facebook, is designed to deliver accurate forecasts for time series data exhibiting natural seasonal patterns, trend shifts, and missing values. Prophet's main strength is its combination of ease of use with a non-parametric and adaptable forecasting structure, setting it apart from conventional time series modeling techniques. The model decomposes the time series into three primary components: trend, seasonality, and holiday or abrupt event effects. These components can be expressed using the Eq. (8):

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t \quad (8)$$

Here, $g(t)$ denotes the non-periodic trend component, which may be linear or saturation-based; $s(t)$ represents one or more periodic seasonal components (e.g., daily, weekly, annual); $h(t)$ corresponds to piecewise functions capturing the effects of holidays and exceptional events; and ϵ_t is the error term, generally assumed to be white noise.

Prophet is capable of modeling the trend function as either a piecewise linear trajectory or a logistic growth curve. Changepoints may be identified automatically or explicitly defined by the analyst. Seasonal variations are modeled using Fourier series expansions.

$$s(t) = \sum_{n=1}^N \left[a_n \cos\left(\frac{2\pi n t}{P}\right) + b_n \sin\left(\frac{2\pi n t}{P}\right) \right] \quad (9)$$

Here, P represents the seasonal period and the number of harmonics. This formulation enables Prophet to offer a periodic structure that is both simple to learn and highly flexible. Holidays and extraordinary events can be directly integrated into the model by providing their date ranges, allowing known calendar shocks to be incorporated in advance.

Prophet is robust to missing observations, suitable for long-term forecasting, capable of decomposing time series into non-linear components, and offers a non-parametric, flexible structure. For these reasons, it was selected as an effective preliminary model for environmental time series, such as air quality, where both trend and seasonality coexist.

Long short-term memory (LSTM)

Long Short-Term Memory (LSTM), an advanced variant of the Recurrent Neural Network (RNN) framework, is widely recognized for its exceptional capability in capturing and modeling long-term dependencies. Conventional RNN architectures exhibit limitations in modeling long-term dependencies in sequential datasets, whereas LSTM networks mitigate this problem through the integration of a persistent cell state and dynamic gating mechanisms. LSTM networks are a specialized type of recurrent neural network designed to overcome challenges such as vanishing and exploding gradients commonly encountered in conventional RNN architectures. Originally proposed by Hochreiter and Schmidhuber in 1997⁴⁵, LSTM has demonstrated exceptional performance in domains such as time series analysis, natural language processing (NLP), and signal analysis, owing to its ability to learn and retain long-term dependencies.

Distinct from traditional RNNs, the LSTM cell incorporates an internal cell state (C_t) that preserves information over extended sequences, ensuring that data dependencies are maintained without degradation. As depicted in Fig. 9, an LSTM cell is structured around three primary gating mechanisms.

Forget gate (f_t) Within this component, a decision is made regarding how much of the previous cell's information should be preserved.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (10)$$

Input Gate (i_t) and candidate cell state (C_t) At this stage, the gating mechanism specifies the degree to which newly acquired input is assimilated into the internal cell state, thereby shaping the model's updated memory representation.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (11)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (12)$$

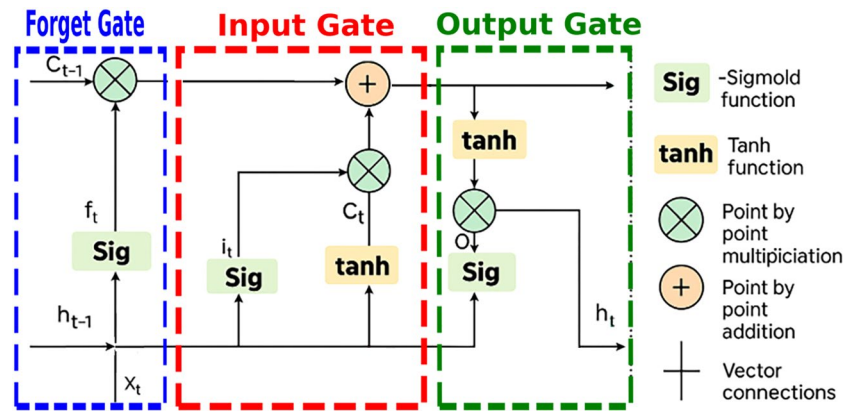


Fig. 9. Structural representation of the LSTM model (adapted from Vignes et al.⁴⁶).

Output gate (o_t) This mechanism governs the extent to which the cell state's contents are exposed at the output.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (13)$$

$$h_t = o_t \cdot \tanh(C_t) \quad (14)$$

The cell state C_t is updated, as illustrated in Eq. (15), by the integration of the forget and input gates.

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (15)$$

Residual-enhanced hybrid forecasting model: Prophet + LSTM

While statistical models, such as Facebook Prophet, have demonstrated strong performance in modeling trends, seasonality, and holiday effects, they often fall short in capturing the complex nonlinear dependencies and abrupt fluctuations present in real-world data. To address this limitation, this study proposes a novel residual-enhanced hybrid forecasting approach that combines the Prophet with an LSTM neural network. In the proposed method, Prophet is first used to generate baseline forecasts and corresponding residuals. These residuals, which represent the unexplained variance in the time series, are then modeled by an LSTM network to capture underlying nonlinear patterns. During inference, the final forecast is obtained by augmenting the Prophet prediction with the LSTM-estimated residuals. This two-stage framework leverages the interpretability and structure of Prophet with the learning capacity of deep neural networks, resulting in improved predictive accuracy across complex time series datasets.

As shown in Fig. 10, proposed method represents a three-phase hybrid time series framework in which the residuals generated by the classical Prophet model are learned by an LSTM network, with the aim of improving its overall predictive accuracy. Initially, the Prophet model is executed using the training dataset. The residual sequences obtained from this stage are modeled by an LSTM network to capture temporal error dependencies. Finally, the Prophet-generated forecasts for the test dataset are refined by incorporating the residual corrections predicted by the LSTM.

Flowchart illustrating the proposed hybrid forecasting framework. The time series data undergoes preprocessing and is split into training (80%) and testing (20%) sets. The Prophet model is trained on the training data to generate initial predictions, and the residuals are calculated on the test set. These residuals are then split again, with 80% used to train an LSTM model and 20% reserved for testing. The LSTM's residual predictions are combined with the Prophet's predictions to produce the hybrid forecast, which is then evaluated using MAE and MSE metrics.

Baseline forecasting using Prophet The training data is processed through the Prophet model, yielding both predicted values and their associated residuals. The baseline prediction $\hat{y}_t^{(P)}$ is computed from the training data. The residuals are then defined as:

$$e_t = y_t - \hat{y}_t^{(P)}, \quad \text{for } t \in \text{Training set} \quad (16)$$

Modeling residuals using LSTM The residual sequence e_t is assumed to contain nonlinear temporal dependencies not captured by Prophet. These residuals are modeled using an LSTM network as:

$$\hat{e}_t = \mathcal{F}_{\text{LSTM}}(e_{t-1}, e_{t-2}, \dots, e_{t-n}) \quad (17)$$

Here, $\mathcal{F}_{\text{LSTM}}$ denotes the learned nonlinear mapping by the LSTM, and n is the input window size for sequence learning. The LSTM is trained to minimize the residual loss $\mathcal{L}_{\text{LSTM}}$.

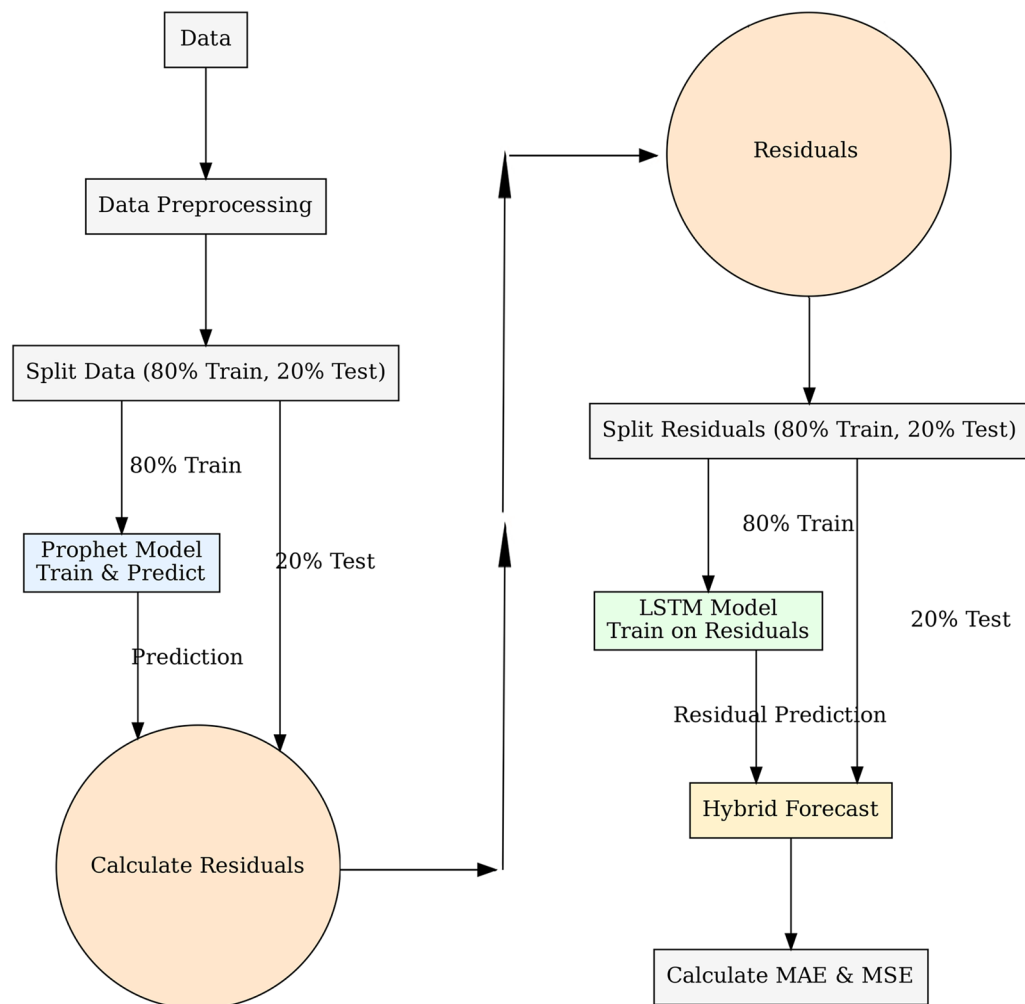


Fig. 10. Proposed hybrid forecasting framework using Prophet and LSTM.

$$\mathcal{L}_{\text{LSTM}} = \frac{1}{N} \sum_{t=n+1}^T (e_t - \hat{e}_t)^2 \quad (18)$$

Final forecast composition During inference, for unseen test data x_t , Prophet provides a primary forecast using Equation 16. Simultaneously, the LSTM predicts the error term $\hat{e}_t$ based on recent residual dynamics. The final hybrid forecast $\hat{y}_t^{(H)}$ is computed as:

$$\hat{y}_t^{(H)} = \hat{y}_t^{(P)} + \hat{e}_t \quad (19)$$

Thus, the model effectively corrects Prophet's systematic errors using LSTM's learned representation of residual structure.

The proposed hybrid forecasting framework offers several distinct advantages by integrating the strengths of both statistical and deep learning methodologies. Firstly, using the Prophet model in the initial stage provides high interpretability, as its components explicitly decompose the time series into trend, seasonality, and holiday effects. This transparency helps domain experts to better understand the temporal dynamics and supports informed decision-making. At the next stage, the task of modelling residual errors is delegated to an LSTM network, enabling the model to provide an effective mechanism for error compensation. The LSTM has the capacity to identify systematic errors in Prophet's forecasts and to gradually learn both their structure and magnitude over time. This enables the model to correct chronic biases in forecasting and adjust to complex nonlinear patterns that traditional methods often fail to capture. Moreover, the proposed framework adopts a modular design, allowing each stage (Prophet and LSTM) to be trained and fine-tuned independently. This separation not only simplifies model development and debugging but also enables flexible adaptation to varying data characteristics, making the approach both robust and reliable across diverse time series forecasting tasks.

Evaluation measures

To rigorously measure the predictive capability of the proposed methodology and to facilitate comparative analysis with alternative forecasting models, two canonical error metrics (Mean Absolute Error (MAE) and Root Mean Square Error (RMSE)) were utilized. These metrics are commonly recognized in the literature as standard measures of predictive accuracy in time series forecasting.

MAE computes the arithmetic mean of the absolute differences between the predicted and actual values, directly reflecting the magnitude of errors. It assumes that each error has equal importance. Since it is expressed in the same units as the data, it can be interpreted directly.

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t| \tag{20}$$

RMSE is the square root of the mean of the squared errors. It penalizes large errors more heavily, as the errors are squared before averaging. Compared to MAE, RMSE is more sensitive, and it can be higher when outliers are present. The units are identical to the original values, and it is commonly used to assess the variability in predictions.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2} \tag{21}$$

Experimental results and discussion

All experiments were conducted using the Google Colab cloud environment. While statistical models such as Prophet and conventional machine learning algorithms (Random Forest Regressor, XGBoost, Gradient Boosting Regressor, Support Vector Regression, and Voting Regressor) were trained using CPU, deep learning models (LSTM, 1D-CNN, etc.) were executed on Google’s Tesla T4 GPU. Detailed specifications of the experimental runtime environment are summarized in Table 4.

The experimental evaluation in this study is organized into two main parts to ensure clarity and fairness in model comparison. (i) The first part presents a performance comparison among traditional machine learning and statistical models (e.g., Random Forest, Gradient Boosting, and XGBoost) to establish a baseline reference. (ii) The second part compares conventional deep learning models (e.g., LSTM, GRU, and CNN-LSTM), state-of-the-art hybrid approaches from the literature, and the proposed hybrid model developed in this study. This two-stage evaluation structure enables a transparent and comprehensive assessment of the proposed model’s performance across both conventional and advanced forecasting paradigms.

Figure 11 presents a comparative analysis of the evaluated regression models across four performance metrics: MAE, RMSE, training time, and prediction time. The visualization clearly highlights the trade-offs between predictive accuracy and computational efficiency for each method.

In terms of accuracy, the Extra Trees model achieved the lowest RMSE (18.68) and was nearly tied with Random Forest for the lowest MAE (11.25 vs. 11.28), indicating high predictive performance. Random Forest also delivered competitive error values, although it exhibited the longest training time (554.16 s) among all models. XGBoost maintained moderate accuracy, with slightly higher errors than Extra Trees and Random Forest, but excelled in efficiency with extremely fast training (2.74 s) and prediction (0.17 s) times, making it a strong candidate for real-time or resource-constrained applications.

The Gradient Boosting model produced the highest MAE (15.23) and RMSE (23.37), suggesting lower predictive accuracy relative to the others, but offered a very short prediction time (0.08 s), which can be advantageous in time-critical use cases. Bagging Regressor yielded moderate error levels while maintaining a balanced trade-off between training duration (52.98 s) and prediction speed (0.49 s) compared to Random Forest.

From a computational standpoint, XGBoost stands out as the most efficient in both training and inference, whereas Random Forest and Extra Trees require significantly longer training times. However, their superior

Specification	Detail
Platform	Google Colab
CPU	Intel(R) Xeon(R) @ 2.00GHz – 2 virtual cores
Architecture	x86_64 (64-bit)
RAM	13.3 GB
GPU (for deep learning)	NVIDIA Tesla T4 with 15 GB memory
Python version	3.11.12
Key Library Versions	Prophet1.1.6, scikit-learn1.6.1, tensorflow2.18.0
CUDA/driver	CUDA 12.4, NVIDIA Driver 550.54.15
Operating system	Virtualized Ubuntu-based Linux (Colab VM)

Table 4. Hardware and software configuration of the cloud-based runtime environment used for model training.

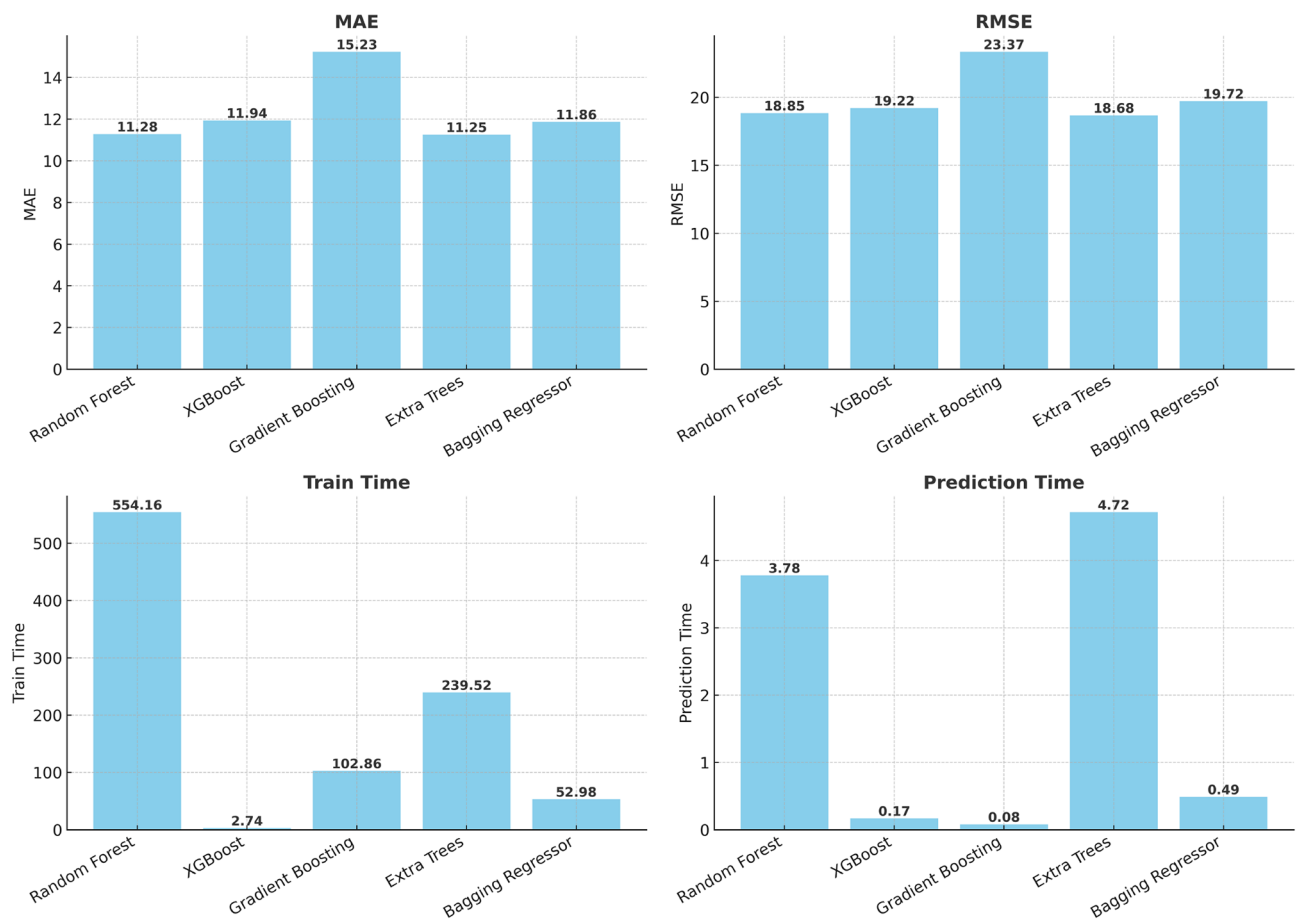


Fig. 11. Comparative performance of five ML models evaluated on MAE, RMSE, training time, and prediction time.

accuracy may justify this cost in contexts where precision is prioritized over speed. Overall, the results emphasize a clear trade-off: Extra Trees and Random Forest lead in predictive accuracy, while XGBoost offers an attractive balance between speed and acceptable error levels.

Figure 12 shows the performance comparison of statistical models before adding additional regressors. As seen, both Prophet and SARIMA models exhibit relatively high MAE and RMSE values, indicating that their baseline forecasting capabilities for $PM_{2.5}$ are limited. To enhance their predictive accuracy, we decided to incorporate additional regressors into the models.

The selection of these regressors was guided by the correlation heatmap presented in Fig. 13. This heatmap illustrates the pairwise Pearson correlation coefficients between $PM_{2.5}$ and other features in the dataset. Based on this analysis, the three features most strongly correlated with $PM_{2.5}$ were identified as PM_{10} ($r = 0.84$), CO ($r = 0.73$), and NO_2 ($r = 0.63$). These variables are known to be closely related to air quality and pollution sources, making them suitable candidates for improving the model's input space.

By integrating these high-correlation variables as external regressors, the models can better capture relationships between pollutant levels and $PM_{2.5}$ concentration, ultimately leading to improved forecasting performance. This regression-enhanced approach is particularly beneficial for statistical models, which rely heavily on informative input features to accurately represent underlying temporal patterns and dependencies.

Figure 14 presents the actual $PM_{2.5}$ values alongside the predictions generated by the baseline Prophet model. The green line represents the actual $PM_{2.5}$ values, while the red line indicates the predicted values. Prophet not only generates a single point forecast but also computes a probability distribution at a given confidence level. The shaded grey area in the figure denotes the uncertainty interval generated by the Prophet model for its forecasts. As can be observed in Fig. 14, the Prophet algorithm succeeds in fitting the actual $PM_{2.5}$ values within the specified confidence bounds; however, its forecasted values remain largely constrained around the mean. This finding suggests that the Prophet model demonstrates strong capability in detecting trend shifts; however, it fails to deliver accurate estimates for $PM_{2.5}$ values with minor deviations from the mean, thereby exhibiting a systematic bias toward the average. Within the developed method, the LSTM model is tasked with learning the systematic bias inherent in Prophet's predictions, which is then added to its future forecasts, resulting in enhanced predictive accuracy.

To assess the empirical sensitivity of the preprocessing parameters, additional experiments were conducted on both the smoothing window and the Z-score threshold. Figure 15 presents the effect of varying the moving

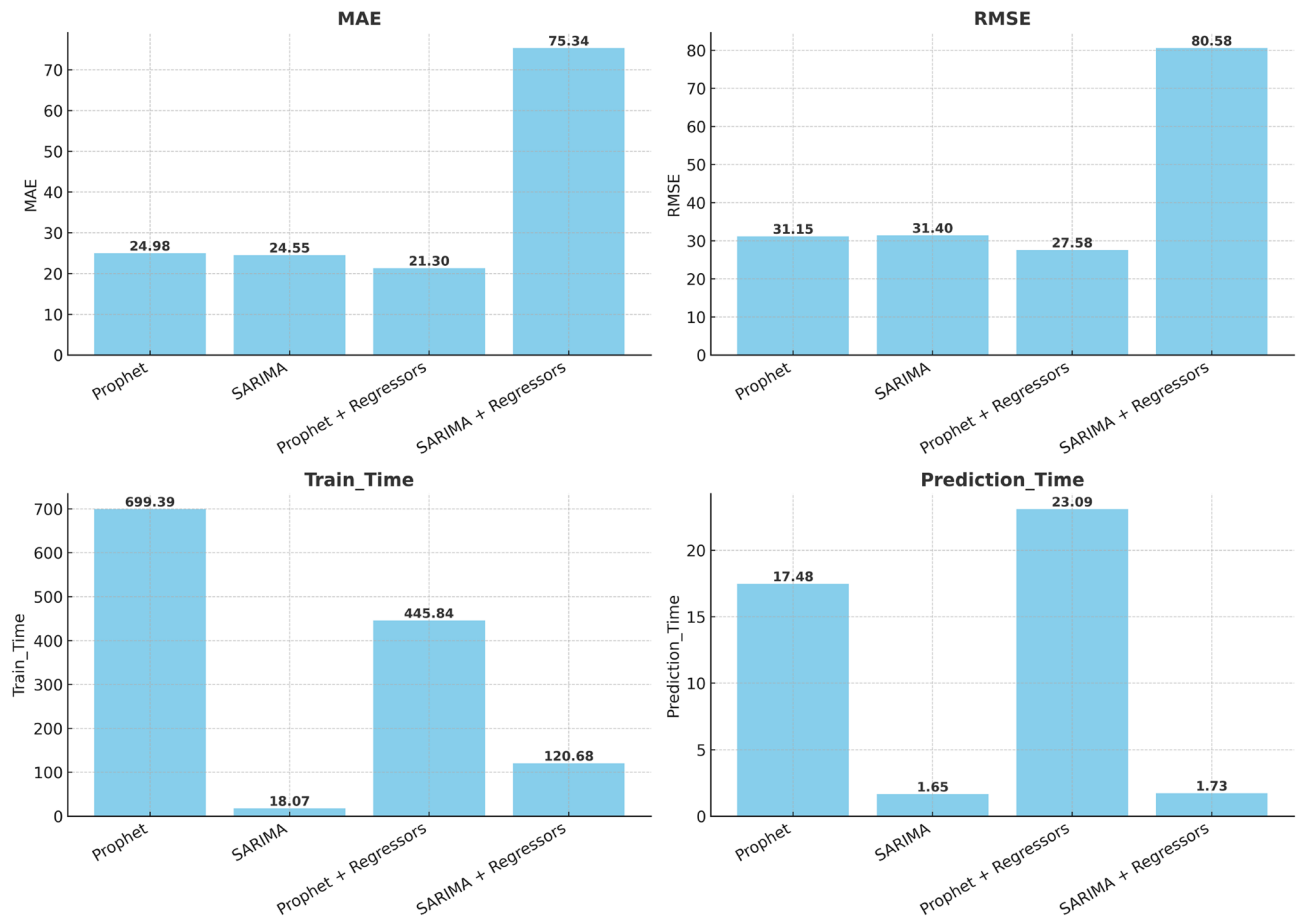


Fig. 12. Performance comparison of statistical forecasting models (Prophet, SARIMA) and their regressor-augmented variants in terms of MAE, RMSE, training time, and prediction time.

average window between 10 and 100 hours on model accuracy. As shown, the overall error metrics (MAE and RMSE) decrease as the window length increases, reaching a stable minimum around the 100-hour setting, which offers the most effective compromise between short-term fluctuation suppression and long-term trend preservation. Figure 16 illustrates the influence of different Z-score thresholds on both forecasting accuracy and data retention. Lower thresholds remove excessive data and distort the distribution, while higher thresholds retain too many outliers and increase error. The threshold of 3.0 demonstrates the optimal balance, removing approximately 1.8% of the samples while maintaining representative high-pollution episodes and ensuring stable model performance.

To quantify the statistical uncertainty of the proposed model's performance, a bootstrap resampling technique was applied directly to the test predictions and their corresponding ground-truth values. Specifically, 1000 bootstrap iterations were performed, each involving random resampling (with replacement) from the test set to estimate the variability of the evaluation metrics without retraining the model. From these resampled distributions, the 95% confidence intervals (CIs) were computed as follows: MAE = 5.04 ± 0.33 (95% CI: [4.73, 5.38]) and RMSE = 8.81 ± 0.67 (95% CI: [8.18, 9.51]). As shown in Fig. 17, the bootstrap distributions exhibit narrow variance, confirming the statistical robustness and consistency of the hybrid Prophet-LSTM model's performance under repeated resampling.

Figure 18 illustrates the comparative analysis between the actual PM_{2.5} concentrations and the forecasts generated by the Prophet model and the hybrid Prophet+LSTM model. As shown in the figure, Prophet exhibits strong capability in detecting trend transitions, but its predictions diverge from the actual observations when sudden spikes or drops in PM_{2.5} values are encountered. This failure emerges as a systematic bias, reflecting the Prophet model's inherent dependence on mean-centered forecasts. Compared to the pure Prophet model, the combined Prophet+LSTM approach exhibits a much tighter correspondence with the actual values, successfully accounting not only for the trend direction but also for the magnitude of fluctuations. Hence, the developed hybrid system synthesizes Prophet's proficiency in capturing local trend shifts with LSTM's robustness in modeling long-range temporal structures and generalization.

As shown in Fig. 18, the standalone Prophet model captures the overall trend and seasonal patterns of PM_{2.5} concentrations effectively but tends to stay close to the short-term moving average, struggling to follow abrupt changes or extreme fluctuations. In contrast, the LSTM refinement successfully addresses these limitations by learning from both the residuals generated by Prophet and the additional spatial indicator feature representing

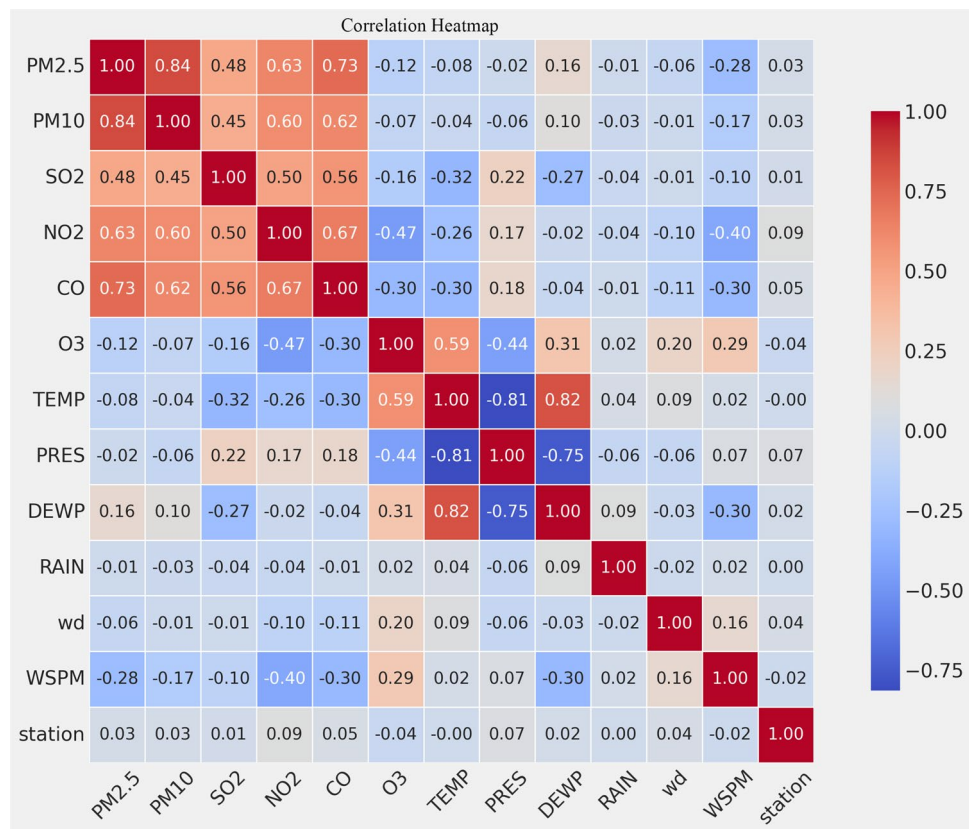


Fig. 13. Correlation heatmap illustrating the relationships between PM_{2.5} concentrations and other features in the dataset. Strong positive correlations are observed between PM_{2.5} and PM₁₀, CO, and NO₂, which informed the selection of these variables as additional regressors for statistical models.

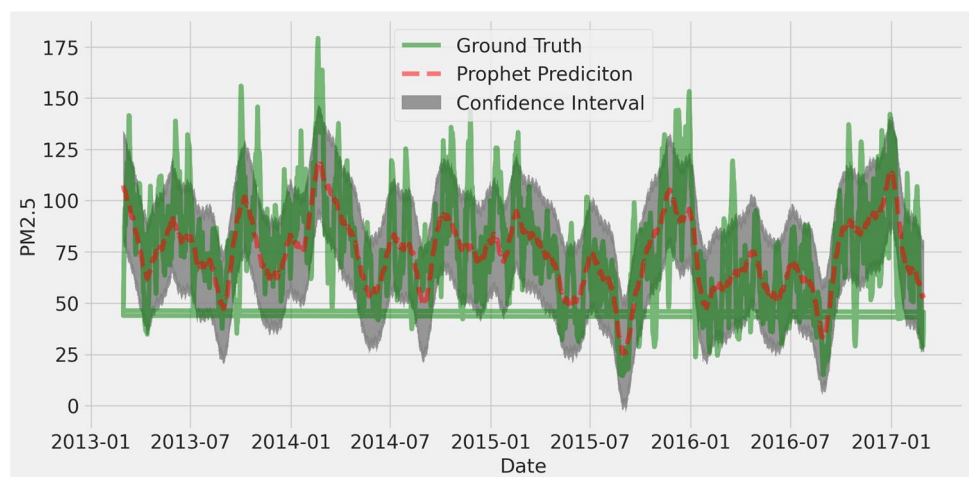


Fig. 14. PM_{2.5} time series forecasting using the Prophet model. The red dashed line represents the model's predictions, the green line shows the actual PM_{2.5} values, and the shaded grey area indicates the prediction uncertainty interval.

each monitoring station. Through this mechanism, the hybrid framework captures inter-station variability and spatial heterogeneity while improving the prediction of short-term peaks that Prophet alone cannot detect. Consequently, the Prophet + LSTM configuration achieves more responsive and accurate forecasts, particularly during sudden pollution surges.

Figure 19 illustrates the deviations of both the Prophet model and the proposed hybrid model from the actual values. In the case of the baseline Prophet model, the errors display a broader spread and demonstrate

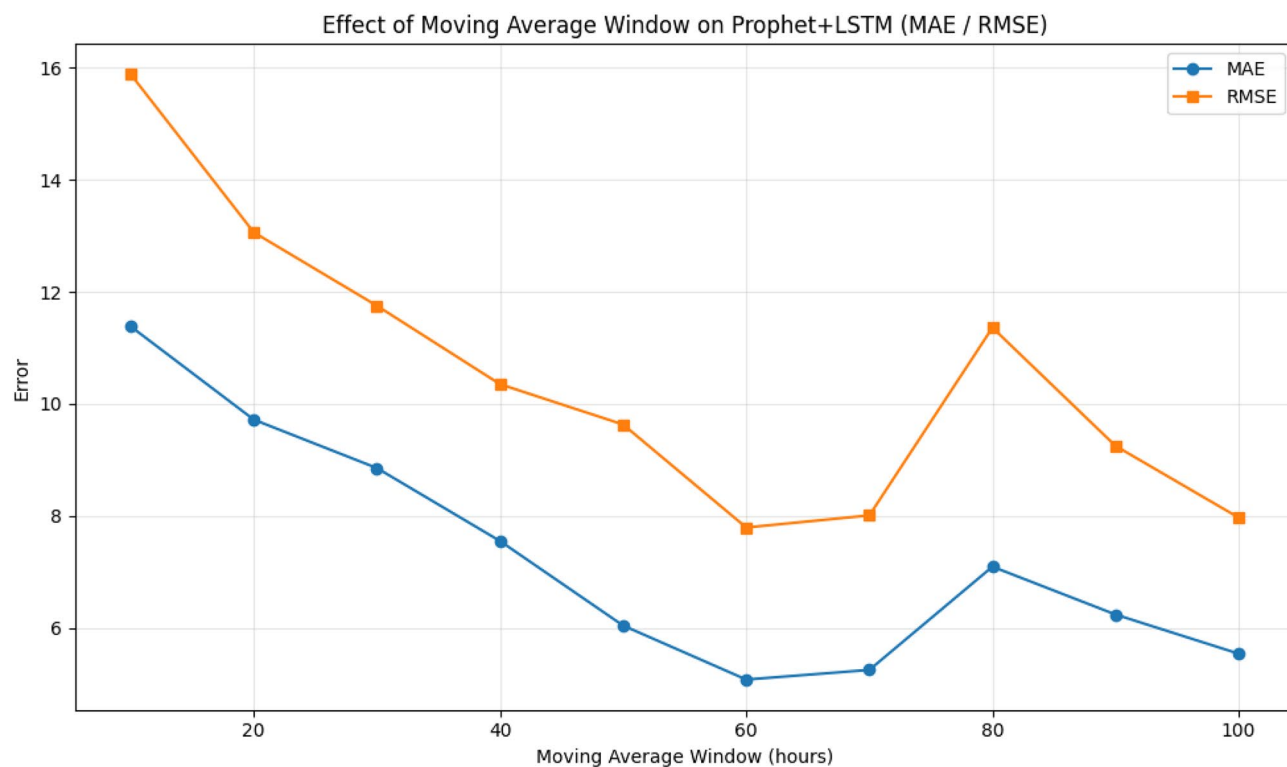


Fig. 15. The impact of different moving average window sizes (10–100 h) on forecasting performance. Both MAE and RMSE values are plotted for each configuration.

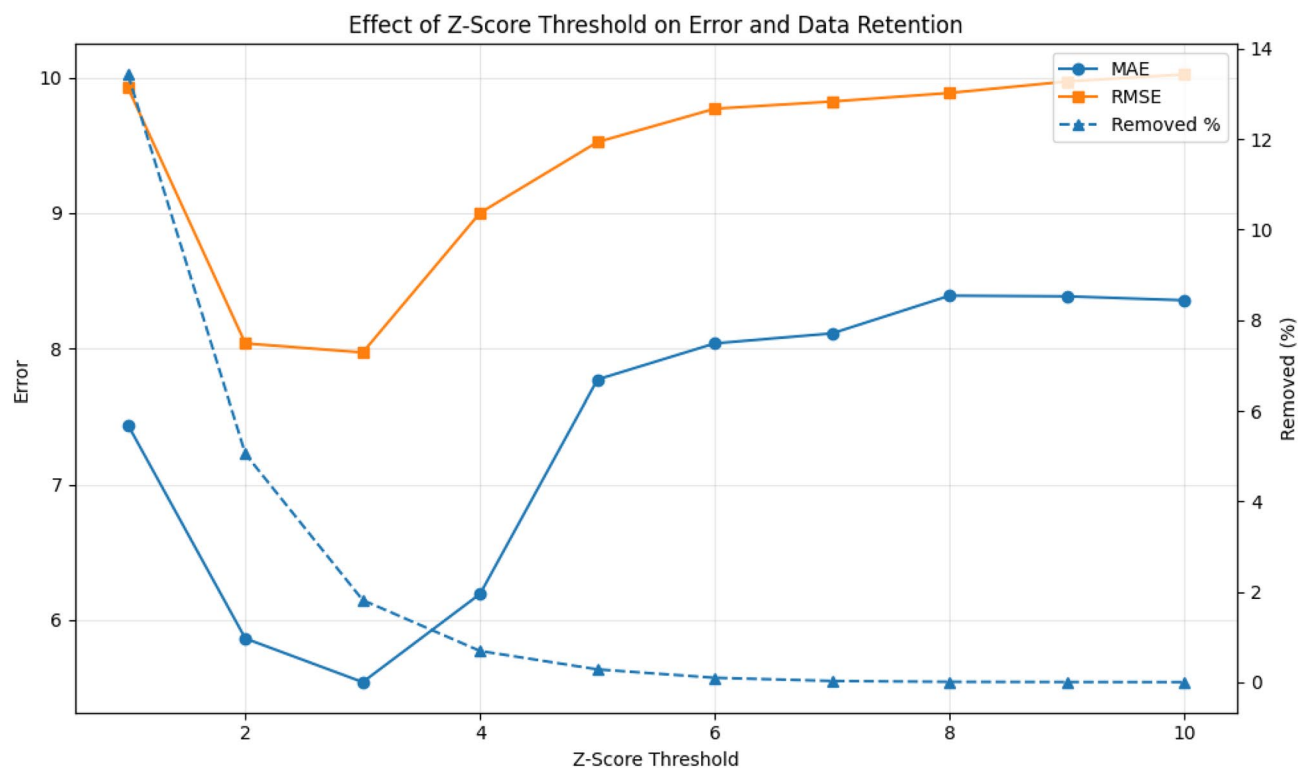


Fig. 16. The relationship between Z-score threshold, forecasting error (MAE/RMSE), and the percentage of removed records.

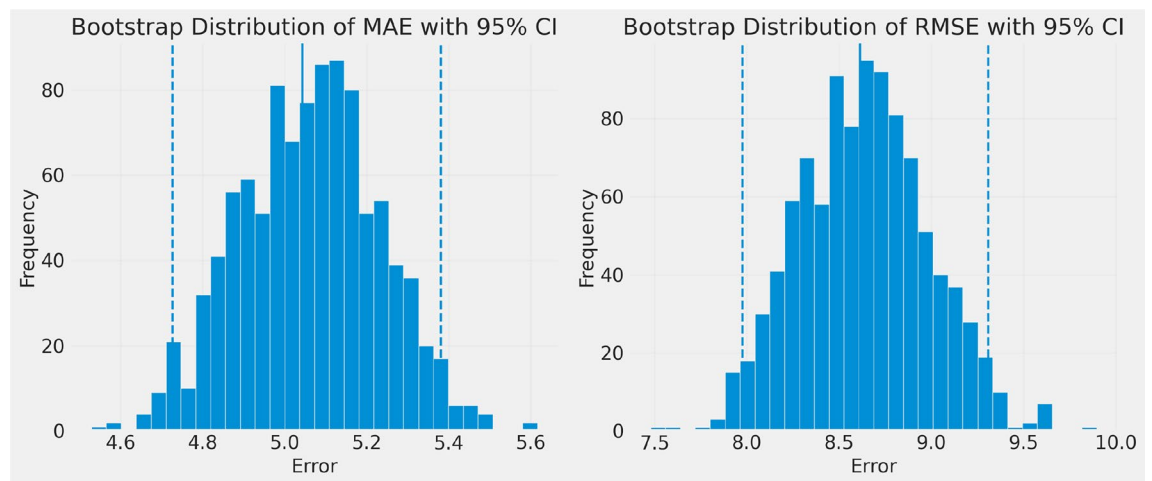


Fig. 17. Bootstrap distributions of MAE and RMSE with 95% confidence intervals, showing the variability and statistical robustness of the hybrid Prophet–LSTM model.

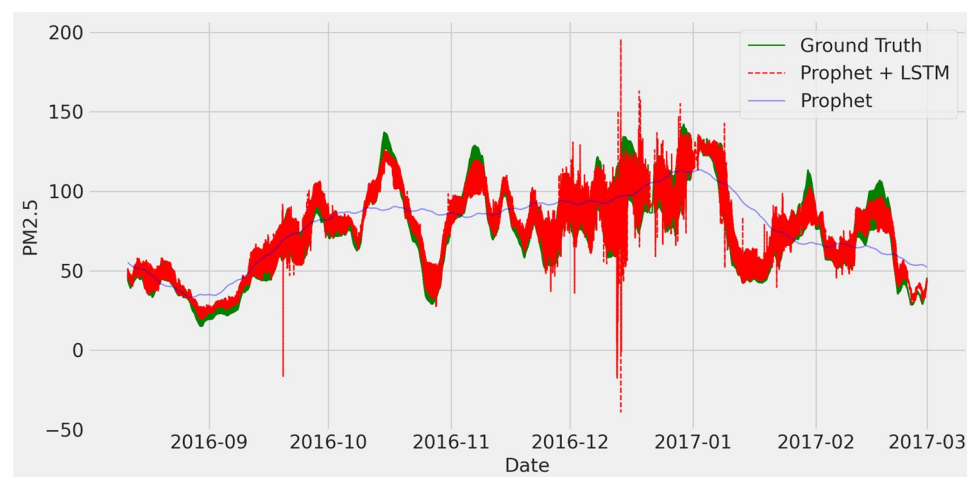


Fig. 18. Comparison of ground truth $PM_{2.5}$ values, baseline Prophet forecasts, and the proposed Prophet+LSTM hybrid model on the test set.

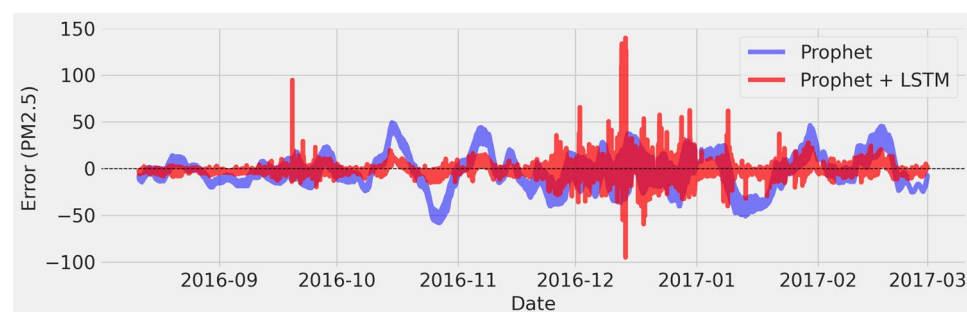


Fig. 19. Error distributions of the Prophet and proposed Prophet+LSTM models over the test set.

clear departures from the zero-error reference line. In contrast, the errors of the proposed Prophet+LSTM model exhibit a narrow distribution around zero, leading to a reduction in the overall magnitude of error. Although isolated spikes remain, particularly during periods of abrupt $PM_{2.5}$ variability, the overall variance is significantly reduced. This finding substantiates the argument that the incorporation of LSTM into the forecasting pipeline

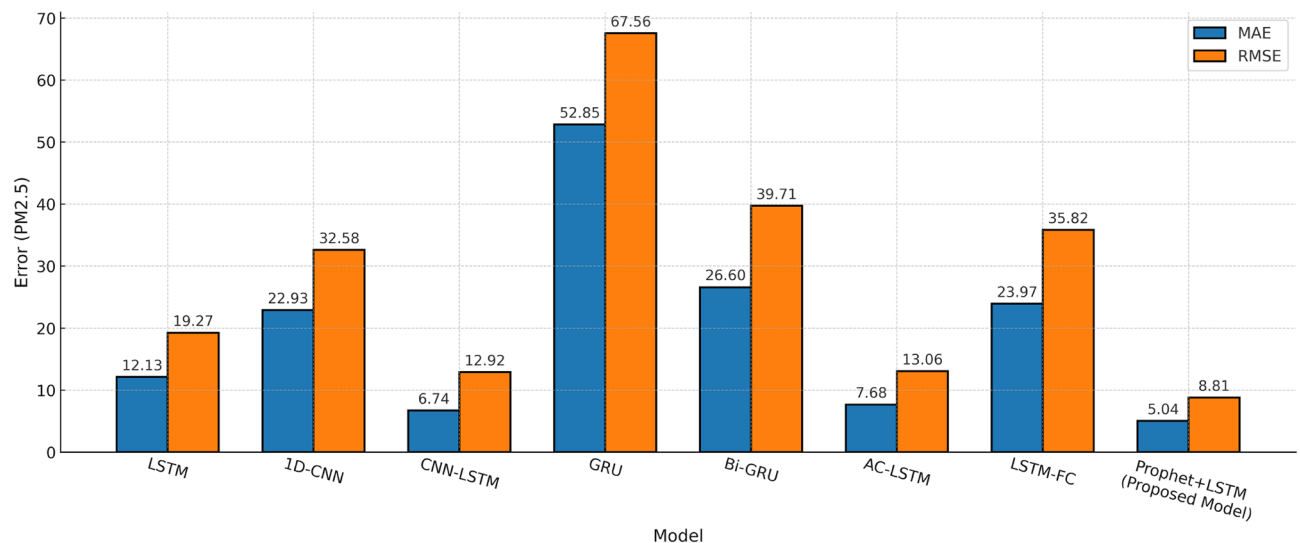


Fig. 20. Comparison of the proposed Prophet+LSTM hybrid model with state-of-the-art deep learning and hybrid approaches in terms of MAE and RMSE.

not only mitigates the limitations of statistical decomposition-based methods but also results in a more robust and reliable predictive framework for air quality time series.

In Fig. 20, the performance of the proposed Prophet+LSTM hybrid model is compared against several state-of-the-art approaches, namely AC-LSTM⁴⁷, LSTM-FC⁴⁸, CNN-LSTM¹⁸, and pure deep learning models in terms of their best MAE and RMSE values for PM_{2.5} prediction. Among the conventional deep learning architectures, the LSTM and CNN-based variants (e.g., CNN-LSTM and AC-LSTM) demonstrate competitive results, indicating their ability to capture temporal dependencies and nonlinear relationships in air quality data. The CNN-LSTM model achieves the lowest error among the standalone deep learning models (MAE = 6.74, RMSE = 12.92), benefiting from the combination of convolutional feature extraction and sequential learning. In contrast, recurrent architectures such as GRU and Bi-GRU exhibit significantly higher error rates, suggesting limited generalization capability under complex temporal dynamics. These findings emphasize the variability in performance across different deep architectures and the importance of hybrid designs in improving predictive accuracy.

When compared with pure DL and hybrid architectures, the proposed Prophet-LSTM model clearly outperforms all competing approaches, achieving the lowest overall error (MAE = 5.04, RMSE = 8.81). This improvement can be attributed to the complementary strengths of the two components: Prophet's ability to accurately and promptly capture trend and seasonal variations in the data, and LSTM's capability to model complex nonlinear patterns in PM_{2.5} concentrations. The integration of these mechanisms allows the model to effectively handle both long-term deterministic changes and short-term irregular fluctuations, resulting in more precise and stable air quality forecasts.

Conclusion

This study proposes a hybrid forecasting framework that leverages the statistical decomposition capability of Prophet and the nonlinear learning power of LSTM networks through residual correction. LSTM significantly improved forecasting accuracy by modeling the systematic residual errors generated by the Prophet algorithm. The experimental findings obtained on the Beijing air quality dataset demonstrated that the Prophet+LSTM model outperformed both traditional statistical techniques and state-of-the-art deep learning models, achieving the lowest MAE and RMSE among all methods. The results validate that the integration of interpretable statistical modeling with residual adjustments learned by deep learning networks yields a stronger and more dependable framework for PM_{2.5} prediction. The developed framework contributes not only methodologically but also operationally, by informing outdoor air quality management and supporting precise, real-time decisions in environmental governance and public health. In future research, the proposed framework can be extended by experimenting with alternative statistical techniques and deep learning models, as well as by integrating real-time sensor networks.

Furthermore, to further assess the generalizability and robustness of the proposed framework, future studies will focus on cross-city and multi-regional validations using air quality datasets from different urban environments. Such evaluations will help determine the model's adaptability to varying emission sources, climatic conditions, and spatial heterogeneity. In addition, extending the framework through domain adaptation or transfer learning techniques could enhance its ability to perform reliably across diverse geographical contexts. These directions will provide a broader validation of the hybrid approach and strengthen its potential for large-scale environmental monitoring and policy applications.

Data availability

The dataset analyzed in this study is publicly available at the UCI Machine Learning Repository (<https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data>).

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References

1. Aguilar-Gomez, S., Dwyer, H., Graff Zivin, J. & Neidell, M. This is air: The “nonhealth” effects of air pollution. *Annu. Rev. Resour. Econ.* **14**, 403–425. <https://doi.org/10.1146/annurev-resource-111820-021816> (2022).
2. Jeong, S. J. The impact of air pollution on human health in Suwon city. *Asian J. Atmos. Environ.* **7**, 227–233. <https://doi.org/10.5572/ajae.2013.7.4.227> (2013).
3. Lelieveld, J., Evans, J. S., Fnais, M., Giannadaki, D. & Pozzer, A. The contribution of outdoor air pollution sources to premature mortality on a global scale. *Nature* **525**, 367–371. <https://doi.org/10.1038/nature15371> (2015).
4. Kelly, F. J. & Fussell, J. C. Air pollution and airway disease. *Clin. Exp. Allergy* **41**, 1059–1071. <https://doi.org/10.1111/j.1365-2222.2011.03776.x> (2011).
5. Elhariri, E. & Taie, S. A. H-ahead multivariate microclimate forecasting system based on deep learning. In *International Conference on Innovative Trends in Computer Engineering (ITCE)*, 168–173. <https://doi.org/10.1109/ITCE.2019.8646540> (Aswan, Egypt, 2019).
6. Ahn, J., Shin, D., Kim, K. & Yang, J. Indoor air quality analysis using deep learning with sensor data. *Sensors* **17**, 2476. <https://doi.org/10.3390/s17112476> (2017).
7. Wu, Y., Qian, C. & Huang, H. Enhanced air quality prediction using a coupled DVMD informer-CNN-LSTM model optimized with dung beetle algorithm. *Entropy* **26**, 534. <https://doi.org/10.3390/e26070534> (2024).
8. Han, Y., Lam, J. C., Li, V. O. & Reiner, D. A Bayesian LSTM model to evaluate the effects of air pollution control regulations in Beijing, China. *Environ. Sci. Policy* **115**, 26–34. <https://doi.org/10.1016/j.envsci.2020.09.017> (2021).
9. Long, H. et al. Revealing long-term indoor air quality prediction: An intelligent informer-based approach. *Sensors* **23**, 8003. <https://doi.org/10.3390/s23188003> (2023).
10. IHME. Global burden of disease (2024)—with minor processing by our world in data: Air pollution (outdoor & indoor). <https://ourworldindata.org/air-pollution> (2024). Accessed 3 Sept 2025.
11. Barker, D. J. The fetal and infant origins of adult disease. *BMJ* **301**, 1111. <https://doi.org/10.1136/bmj.301.6761.1111> (1990).
12. World Health Organization. Who global air quality guidelines: Particulate matter (pm_{2.5} and pm₁₀), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide. <https://www.who.int/publications/i/item/9789240034228> (2021). Accessed 03 Sept 2025.
13. Zhang, X., Zhang, Q. & Zhang, G. A novel hybrid model for air quality prediction based on deep learning. *Atmos. Environ.* **221**, 117145. <https://doi.org/10.1016/j.atmosenv.2019.117145> (2020).
14. Li, X., Peng, L., Hu, Y., Shao, J. & Chi, T. Deep learning architecture for air quality prediction. *Environ. Sci. Pollut. Res.* **26**, 22451–22464. <https://doi.org/10.1007/s11356-016-7812-9> (2019).
15. Hähnel, P., Mareček, J., Monteil, J. & O'Donncha, F. Using deep learning to extend the range of air pollution monitoring and forecasting. *J. Comput. Phys.* **408**, 109278. <https://doi.org/10.1016/j.jcp.2020.109278> (2020).
16. Wang, B., Kong, W., Guan, H. & Xiong, N. N. Air quality forecasting based on gated recurrent long short term memory model in internet of things. *IEEE Access* **7**, 69524–69534. <https://doi.org/10.1109/ACCESS.2019.2917277> (2019).
17. Huang, Y., Xiang, Y., Zhao, R. & Cheng, Z. Air quality prediction using improved PSO-BP neural network. *IEEE Access* **8**, 99346–99353. <https://doi.org/10.1109/ACCESS.2020.2998145> (2020).
18. Bekkar, A., Hssina, B., Douzi, S. & Douzi, K. Air-pollution prediction in smart city, deep learning approach. *J. Big Data* **8**, 161. <https://doi.org/10.1186/s40537-021-00548-1> (2021).
19. Kumar, U. & Jain, V. K. Arima forecasting of ambient air pollutants (o₃, no, no₂ and co). *Stoch. Environ. Res. Risk Assess.* **24**, 751–760. <https://doi.org/10.1007/s00477-009-0361-8> (2010).
20. Shen, J., Valagolam, D. & McCalla, S. Prophet forecasting model: A machine learning approach to predict the concentration of air pollutants (pm_{2.5}, pm₁₀, o₃, no₂, so₂, co) in Seoul, South KoreaK. *PeerJ* **8**, e9961. <https://doi.org/10.7717/peerj.9961> (2020).
21. Xu, Q., Pang, Y. & Liu, Y. Air traffic density prediction using Bayesian ensemble graph attention network (began). *Transp. Res. Part C Emerg. Technol.* **153**, 104225. <https://doi.org/10.1016/j.trc.2023.104225> (2023).
22. Xu, Q., Pang, Y., Zhou, X. & Liu, Y. Pigat: Physics-informed graph attention transformer for air traffic state prediction. *IEEE Trans. Intell. Transp. Syst.* **25**, 12561–12577. <https://doi.org/10.1109/TITS.2024.3386128> (2024).
23. Alsabagh, A. S. et al. Deep learning framework for hourly air pollutants forecasting using encoding cyclical features across multiple monitoring sites in Beijing. *Sci. Rep.* **15**, 22417. <https://doi.org/10.1038/s41598-025-05472-5> (2025).
24. Wang, J., Li, J., Wang, X., Wang, J. & Huang, M. Air quality prediction using CT-LSTM. *Neural Comput. Appl.* **33**, 4779–4792. <https://doi.org/10.1007/s00521-020-05535-w> (2021).
25. Dong, J., Zhang, Y. & Hu, J. Short-term air quality prediction based on EMD-transformer-bilstm. *Sci. Rep.* **14**, 20513. <https://doi.org/10.1038/s41598-024-67626-1> (2024).
26. Yu, C. et al. Mgsformer: A multi-granularity spatiotemporal fusion transformer for air quality prediction. *Inf. Fusion* **113**, 102607. <https://doi.org/10.1016/j.inffus.2024.102607> (2025).
27. Cao, W., Qi, W. & Lu, P. Air quality prediction based on time series decomposition and convolutional sparse self-attention mechanism transformer model. *IEEE Access* **12**, 1–12. <https://doi.org/10.1109/ACCESS.2024.3484579> (2024).
28. Al-Eidi, S. et al. Comparative analysis study for air quality prediction in smart cities using regression techniques. *IEEE Access* **11**, 115140–115149. <https://doi.org/10.1109/ACCESS.2023.3323447> (2023).
29. Ameer, S. et al. Comparative analysis of machine learning techniques for predicting air quality in smart cities. *IEEE Access* **7**, 128325–128338. <https://doi.org/10.1109/ACCESS.2019.2925082> (2019).
30. Zhang, S. et al. Cautionary tales on air-quality improvement in Beijing. *Proc. Roy. Soc. A* **473**, 20170457. <https://doi.org/10.1098/rspa.2017.0457> (2017).
31. Rao, W., Li, J. & Guo, H. Air quality assessment in Beijing based on cloud model. *Sci. Rep.* **15**, 21994. <https://doi.org/10.1038/s41598-025-05751-1> (2025).
32. Hasan, M. K. et al. Missing value imputation affects the performance of machine learning: A review and analysis of the literature (2010–2021). *Inf. Med. Unlocked* **27**, 100799. <https://doi.org/10.1016/j.imu.2021.100799> (2021).
33. Nassif, A. B., Talib, M. A., Nasir, Q. & Dakalbab, F. M. Machine learning for anomaly detection: A systematic review. *IEEE Access* **9**, 78658–78700. <https://doi.org/10.1109/ACCESS.2021.3083060> (2021).
34. Aggarwal, C. C. *An Introduction to Outlier Analysis* Vol. 22 (Springer Publishing, Cham, 2017).
35. Saleem, S., Aslam, M. & Shaukat, M. R. A review and empirical comparison of univariate outlier detection methods. *Pak. J. Statist.* **37**, 447–462 (2021).
36. Rodríguez, P., Bautista, M. A., Gonzalez, J. & Escalera, S. Beyond one-hot encoding: Lower dimensional target embedding. *Image Vis. Comput.* **75**, 21–31. <https://doi.org/10.1016/j.imavis.2018.04.004> (2018).

37. Dahouda, M. K. & Joe, I. A deep-learned embedding technique for categorical features encoding. *IEEE Access* **9**, 114381–114391. <https://doi.org/10.1109/ACCESS.2021.3104357> (2021).
38. Weiss, S. et al. Normalization and microbial differential abundance strategies depend upon data characteristics. *Microbiome* **5**, 27. <https://doi.org/10.1186/s40168-017-0237-y> (2017).
39. Passalis, N., Tefas, A., Kannianen, J., Gabbouj, M. & Iosifidis, A. Deep adaptive input normalization for time series forecasting. *IEEE Trans. Neural Netw. Learn. Syst.* **31**, 3760–3765. <https://doi.org/10.1109/TNNLS.2019.2944933> (2019).
40. Milli, M. & Bulut, H. Substream: Online subtractive stream clustering algorithm. *Concurrency Comput. Pract. Exper.* **34**, e6968. <https://doi.org/10.1002/cpe.6968> (2022).
41. Aggarwal, C. C. Outlier ensembles. In *Outlier Analysis*, 185–218, https://doi.org/10.1007/978-3-319-47578-3_6 (Springer Int. Publishing, Cham, Switzerland, 2016), 2nd edn.
42. Tian, Y. & Zhang, Y. A comprehensive survey on regularization strategies in machine learning. *Inf. Fusion* **80**, 146–166. <https://doi.org/10.1016/j.inffus.2021.11.005> (2022).
43. Zhu, D., Cai, C., Yang, T. & Zhou, X. A machine learning approach for air quality prediction: Model regularization and optimization. *Big Data Cogn. Comput.* **2**, 5. <https://doi.org/10.3390/bdcc2010005> (2018).
44. Ventura, L. M. B., de Oliveira Pinto, F., Soares, L. M., Luna, A. S. & Gioda, A. Forecast of daily pm2.5 concentrations applying artificial neural networks and holt-winters models. *Air Qual. Atmos. Health* **12**, 317–325. <https://doi.org/10.1007/s11869-018-00660-x> (2019).
45. Hochreiter, S. & Schmidhuber, J. Long short-term memory. *Neural Comput.* **9**, 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735> (1997).
46. Vm, V., Mp, S. H., Satheesh, R., Das, V. & Padmanaban, S. Ai-driven cybersecurity framework for anomaly detection in power systems: Vignes vm et al.. *Sci. Rep.* **15**, 35506. <https://doi.org/10.1038/s41598-025-19634-y> (2025).
47. Li, S. et al. Urban pm2.5 concentration prediction via attention-based CNN-LSTM. *Appl. Sci.* **10**, 1953. <https://doi.org/10.3390/ap10061953> (2020).
48. Zhao, J., Deng, F., Cai, Y. & Chen, J. Long short-term memory-fully connected (LSTM-FC) neural network for pm2.5 concentration prediction. *Chemosphere* **220**, 486–492. <https://doi.org/10.1016/j.chemosphere.2018.12.128> (2019).

Author contributions

All aspects of this study, including design, implementation, analysis, and writing, were performed by M.M.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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